STAT3888: Nonlinear Relationships Analysis

AUTHOR
Calvin Lai

PUBLISHED
September 16, 2026

To determine if there are possible nonlinear relationships, we will consider fitting *decision trees* and *random forests*. These are mostly meant to be supplementary analysis and likely will not completely replace our multinomial logistic regression as the main model for association and inference.

As for generalised additive models, we may not want to use them for now since all our variables are categorical anyway so there really isn't anything to smooth.

Classification Decision Tree

To start the tree analysis, we use a simple decision tree using `rpart`. We will mostly just use the full data set for now.

▼ Code

```
# Load data set
fdat.full = readRDS(here("Objects", "fdat_full.rds")) %>% drop_
# Decision tree fit for ALT; need cp very small for something i
res.alt = rpart(ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB
            control = rpart.control(cp = 0))
res.alt
```

```
n= 5340

node), split, n, loss, yval, (yprob)
      * denotes terminal node

   1) root 5340 1228 Low (0.77003745 0.11011236 0.11985019)
     2) WAIST=Increased risk,Not at risk 2887  479 Low
(0.83408382 0.09109803 0.07481815) *
     3) WAIST=Substantially increased risk 2453  749 Low
(0.69465960 0.13249083 0.17284957)
       6) AGEB=20-24 years,70-74 years,75-79 years,80-84
years,85 years and older 503   73 Low (0.85487078 0.07952286
0.06560636) *
       7) AGEB=18-19 years,25-29 years,30-34 years,35-39
```

```
years,40-44 years,45-49 years,50-54 years,55-59 years,60-64
years,65-69 years 1950  676 Low (0.65333333 0.14615385
0.20051282)
        14) SEX=Female 1177  310 Low (0.73661852 0.10025489
0.16312659) *
        15) SEX=Male 773  366 Low (0.52652005 0.21604140
0.25743855)
          30) AGEB=55-59 years,60-64 years,65-69 years 364  126
Low (0.65384615 0.18681319 0.15934066) *
          31) AGEB=18-19 years,25-29 years,30-34 years,35-39
years,40-44 years,45-49 years,50-54 years 409  240 Low
(0.41320293 0.24205379 0.34474328)
            62) AGEB=18-19 years,40-44 years 79   40 Low
(0.49367089 0.13924051 0.36708861)
             124) DSMKHHRB=None,Two 58   27 Low (0.53448276
0.15517241 0.31034483) *
             125) DSMKHHRB=One 21   10 High (0.38095238
0.09523810 0.52380952)
               250) EXREGUIC=Met recommended guidelines 10    5
Low (0.50000000 0.10000000 0.40000000) *
               251) EXREGUIC=Did not meet recommended
guidelines 11    4 High (0.27272727 0.09090909 0.63636364) *
            63) AGEB=25-29 years,30-34 years,35-39 years,45-49
years,50-54 years 330  200 Low (0.39393939 0.26666667
0.33939394)
             126) DSMKHHRB=One,Three or more 57   26 Low
(0.54385965 0.19298246 0.26315789)
               252) AGEB=35-39 years,50-54 years 22    6 Low
(0.72727273 0.09090909 0.18181818) *
               253) AGEB=25-29 years,30-34 years,45-49 years 35
20 Low (0.42857143 0.25714286 0.31428571)
                 506) EXREGUIC=Met recommended guidelines 17
8 Low (0.52941176 0.23529412 0.23529412) *
                 507) EXREGUIC=Did not meet recommended
guidelines 18   11 High (0.33333333 0.27777778 0.38888889) *
             127) DSMKHHRB=None,Two 273  174 Low (0.36263736
0.28205128 0.35531136)
               254) AGEB=25-29 years,30-34 years,45-49
years,50-54 years 227  140 Low (0.38325991 0.28193833
0.33480176)
                 508) EXREGUIC=Met recommended guidelines 106
67 Low (0.36792453 0.33018868 0.30188679)
                  1016) AGEB=30-34 years,50-54 years 52   32
Low (0.38461538 0.38461538 0.23076923)
                    2032) AGEB=30-34 years 18    9 Low
(0.50000000 0.33333333 0.16666667) *
                    2033) AGEB=50-54 years 34   20 Borderline
(0.32352941 0.41176471 0.26470588) *
```

```
                 1017) AGEB=25-29 years,45-49 years 54   34
High (0.35185185 0.27777778 0.37037037) *
                509) EXREGUIC=Did not meet recommended
guidelines 121   73 Low (0.39669421 0.23966942 0.36363636)
                 1018) AGEB=25-29 years,45-49 years,50-54
years 100   56 Low (0.44000000 0.23000000 0.33000000) *
                 1019) AGEB=30-34 years 21   10 High
(0.19047619 0.28571429 0.52380952) *
              255) AGEB=35-39 years 46   25 High (0.26086957
0.28260870 0.45652174) *
```

▼ Code

```
printcp(res.alt)
```

```
Classification tree:
rpart(formula = ALT3 ~ EXREGUIC + AGEB + SEX + WAIST +
DSMKHHRB,
    data = fdat.full, control = rpart.control(cp = 0))

Variables actually used in tree construction:
[1] AGEB     DSMKHHRB EXREGUIC SEX      WAIST

Root node error: 1228/5340 = 0.22996

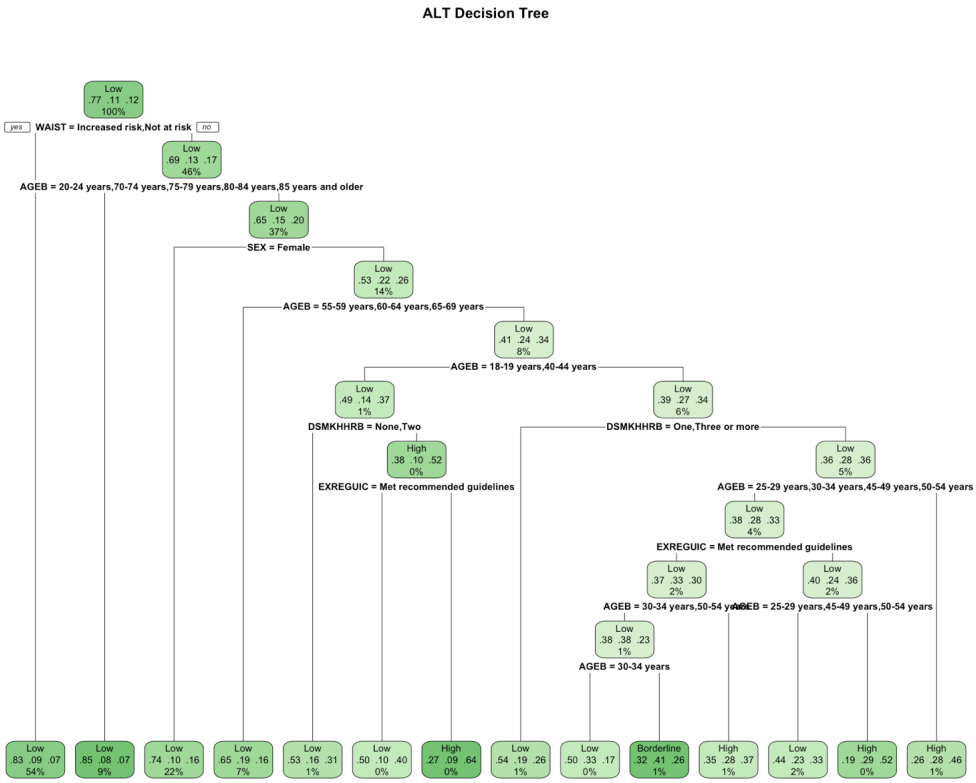
n= 5340

          CP nsplit rel error xerror     xstd
1 0.00122150      0   1.00000 1.0000 0.025041
2 0.00081433     12   0.98127 1.0155 0.025176
3 0.00040717     13   0.98046 1.0147 0.025169
4 0.00000000     15   0.97964 1.0212 0.025225
```

▼ Code

```
res.alt.pruned = prune(res.alt, cp = 0.00081433)

# Plot
rpart.plot(res.alt.pruned, box.palette = "Greens", type = 2, fa
           tweak = 1.2, main = "ALT Decision Tree")
```



▼ Code

```
# Decision tree fit for GGT
res.ggt = rpart(GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,
            control = rpart.control(cp = 0))
res.ggt
```

```
n= 5340

node), split, n, loss, yval, (yprob)
      * denotes terminal node

  1) root 5340 996 Low (0.81348315 0.05505618 0.13146067)
    2) WAIST=Increased risk,Not at risk 2887 361 Low
(0.87495670 0.04191202 0.08313128)
      4) AGEB=18-19 years,20-24 years,25-29 years,30-34
years,35-39 years 914  62 Low (0.93216630 0.03282276
0.03501094) *
      5) AGEB=40-44 years,45-49 years,50-54 years,55-59
years,60-64 years,65-69 years,70-74 years,75-79 years,80-84
years,85 years and older 1973 299 Low (0.84845413 0.04612266
0.10542321)
       10) EXREGUIC=Met recommended guidelines 1143 136 Low
(0.88101487 0.04286964 0.07611549) *
```

```
       11) EXREGUIC=Did not meet recommended guidelines 830 163
Low (0.80361446 0.05060241 0.14578313)
         22) AGEB=40-44 years,45-49 years,50-54 years,55-59
years,65-69 years,70-74 years,75-79 years,80-84 years 716 128
Low (0.82122905 0.04748603 0.13128492) *
         23) AGEB=60-64 years,85 years and older 114  35 Low
(0.69298246 0.07017544 0.23684211)
           46) DSMKHHRB=None 88  24 Low (0.72727273 0.07954545
0.19318182) *
           47) DSMKHHRB=One,Two 26  11 Low (0.57692308
0.03846154 0.38461538)
             94) WAIST=Not at risk 12   3 Low (0.75000000
0.08333333 0.16666667) *
             95) WAIST=Increased risk 14   6 High (0.42857143
0.00000000 0.57142857) *
    3) WAIST=Substantially increased risk 2453 635 Low
(0.74113331 0.07052589 0.18834081)
      6) AGEB=18-19 years,20-24 years,25-29 years,30-34
years,35-39 years,40-44 years,75-79 years,80-84 years,85 years
and older 865 183 Low (0.78843931 0.06127168 0.15028902)
       12) DSMKHHRB=None,Three or more 693 129 Low (0.81385281
0.04906205 0.13708514) *
       13) DSMKHHRB=One,Two 172  54 Low (0.68604651 0.11046512
0.20348837)
         26) SEX=Female 96  23 Low (0.76041667 0.08333333
0.15625000) *
         27) SEX=Male 76  31 Low (0.59210526 0.14473684
0.26315789)
           54) AGEB=18-19 years,25-29 years,40-44 years,75-79
years 44  13 Low (0.70454545 0.04545455 0.25000000) *
           55) AGEB=20-24 years,30-34 years,35-39 years,80-84
years,85 years and older 32  18 Low (0.43750000 0.28125000
0.28125000)
            110) EXREGUIC=Met recommended guidelines 10   4 Low
(0.60000000 0.00000000 0.40000000) *
            111) EXREGUIC=Did not meet recommended guidelines
22  13 Borderline (0.36363636 0.40909091 0.22727273)
              222) AGEB=20-24 years,30-34 years,80-84 years,85
years and older 13   6 Borderline (0.30769231 0.53846154
0.15384615) *
              223) AGEB=35-39 years 9   5 Low (0.44444444
0.22222222 0.33333333) *
      7) AGEB=45-49 years,50-54 years,55-59 years,60-64
years,65-69 years,70-74 years 1588 452 Low (0.71536524
0.07556675 0.20906801)
       14) AGEB=45-49 years,55-59 years,70-74 years 709 190 Low
(0.73201693 0.07898449 0.18899859) *
       15) AGEB=50-54 years,60-64 years,65-69 years 879 262 Low
```

```
(0.70193402 0.07281001 0.22525597)
         30) EXREGUIC=Met recommended guidelines 389 106 Low
(0.72750643 0.06940874 0.20308483) *
         31) EXREGUIC=Did not meet recommended guidelines 490
156 Low (0.68163265 0.07551020 0.24285714)
           62) SEX=Male 193  57 Low (0.70466321 0.09844560
0.19689119)
            124) AGEB=60-64 years,65-69 years 137  36 Low
(0.73722628 0.08029197 0.18248175) *
            125) AGEB=50-54 years 56  21 Low (0.62500000
0.14285714 0.23214286)
              250) DSMKHHRB=None 49  16 Low (0.67346939
0.14285714 0.18367347) *
              251) DSMKHHRB=One,Two 7   3 High (0.28571429
0.14285714 0.57142857) *
           63) SEX=Female 297  99 Low (0.66666667 0.06060606
0.27272727) *
```

▼ Code

```
printcp(res.ggt) # tells us which gives smallest xerror
```

```
Classification tree:
rpart(formula = GGT3 ~ EXREGUIC + AGEB + SEX + WAIST +
DSMKHHRB,
    data = fdat.full, control = rpart.control(cp = 0))

Variables actually used in tree construction:
[1] AGEB     DSMKHHRB EXREGUIC SEX      WAIST

Root node error: 996/5340 = 0.18652

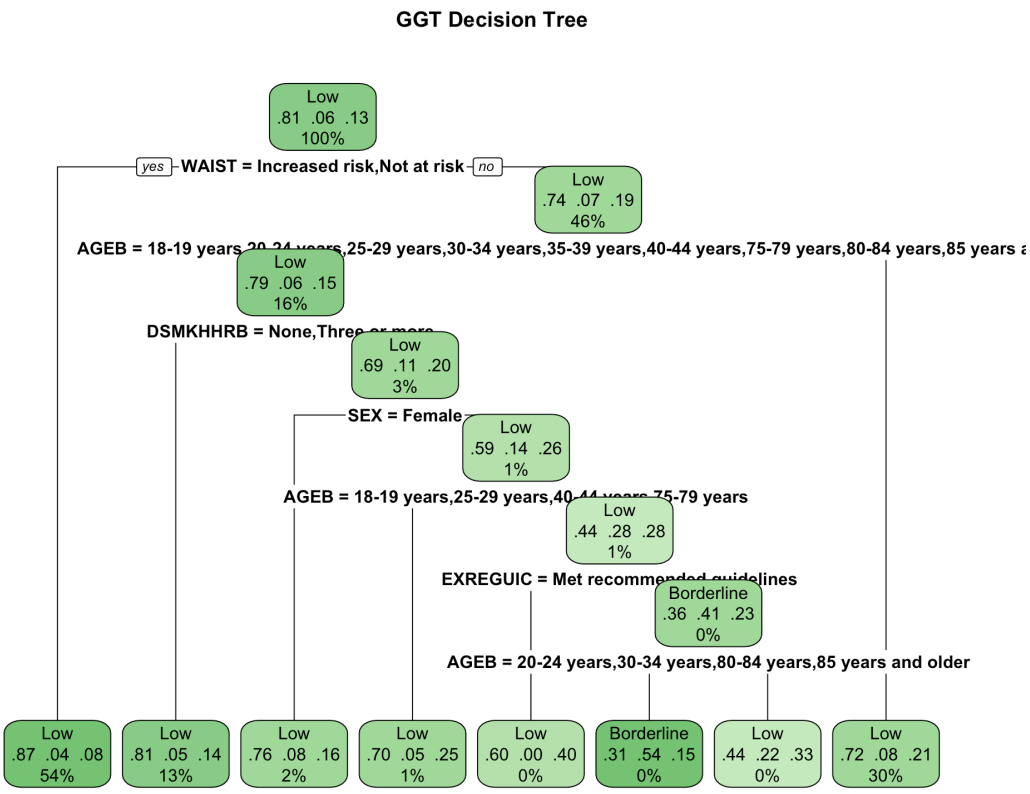
n= 5340

          CP nsplit rel error xerror     xstd
1 0.00043029      0   1.00000 1.0000 0.028579
2 0.00040161      7   0.99699 1.0131 0.028722
3 0.00000000     17   0.99297 1.0151 0.028743
```

▼ Code

```
res.ggt.pruned = prune(res.ggt, cp = 0.00043029)

# Plot
rpart.plot(res.ggt.pruned, box.palette = "Greens", type = 2, fa
           tweak = 1.2, main = "GGT Decision Tree")
```



Cross-Validation

▼ Code

```
# ALT CV
set.seed(3888)
ctrl = trainControl(method = "repeatedcv", number = 10, repeats
                    savePredictions = "final")
tree.cv.alt = train(ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKH
                    data = fdat.full, method = "rpart", trContr
                    tuneLength = 15, metric = "Accuracy")
tree.cv.alt
```

```
CART

5340 samples
   5 predictor
   3 classes: 'Low', 'Borderline', 'High'

No pre-processing
```

```
Resampling: Cross-Validated (10 fold, repeated 3 times)
Summary of sample sizes: 4806, 4806, 4806, 4806, 4806, 4806,
...
Resampling results across tuning parameters:

  cp            Accuracy   Kappa
  0.000000e+00  0.7671661  0.07067970
  4.362494e-05  0.7671661  0.07067970
  8.724988e-05  0.7671661  0.07067970
  1.308748e-04  0.7672911  0.07088227
  1.744998e-04  0.7672911  0.07088227
  2.181247e-04  0.7675408  0.07127377
  2.617497e-04  0.7675408  0.07127377
  3.053746e-04  0.7671662  0.06897923
  3.489995e-04  0.7671662  0.06897923
  3.926245e-04  0.7671662  0.06897923
  4.362494e-04  0.7675404  0.06265870
  4.798744e-04  0.7674159  0.05616991
  5.234993e-04  0.7673535  0.05423532
  5.671242e-04  0.7669165  0.04527588
  6.107492e-04  0.7671033  0.03028188

Accuracy was used to select the optimal model using the largest
value.
The final value used for the model was cp = 0.0002617497.
```

▼ Code

```
tree.cv.alt$bestTune
```

```
           cp
7 0.0002617497
```

▼ Code

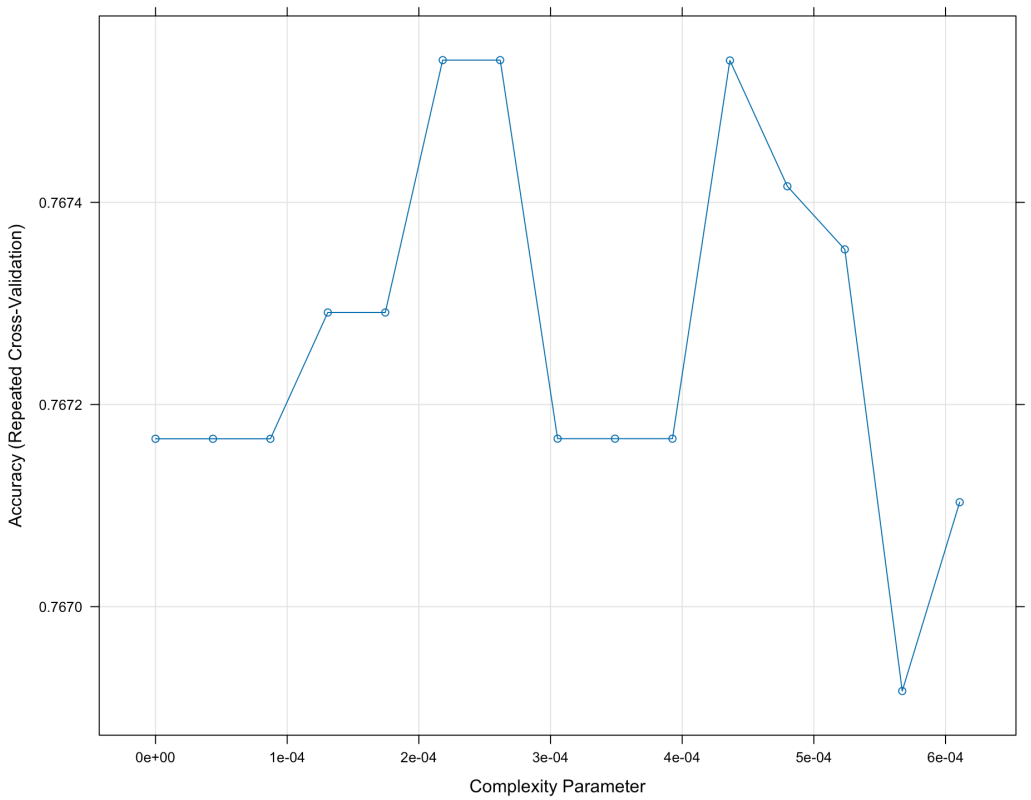
```
tree.cv.alt$results
```

```
             cp  Accuracy      Kappa  AccuracySD    KappaSD
1  0.000000e+00 0.7671661 0.07067970 0.007853756 0.02319142
2  4.362494e-05 0.7671661 0.07067970 0.007853756 0.02319142
3  8.724988e-05 0.7671661 0.07067970 0.007853756 0.02319142
4  1.308748e-04 0.7672911 0.07088227 0.007936110 0.02326709
5  1.744998e-04 0.7672911 0.07088227 0.007936110 0.02326709
6  2.181247e-04 0.7675408 0.07127377 0.007825772 0.02345382
7  2.617497e-04 0.7675408 0.07127377 0.007825772 0.02345382
8  3.053746e-04 0.7671662 0.06897923 0.007911841 0.02418681
9  3.489995e-04 0.7671662 0.06897923 0.007911841 0.02418681
```

```
10 3.926245e-04 0.7671662 0.06897923 0.007911841 0.02418681
11 4.362494e-04 0.7675404 0.06265870 0.007898047 0.02795021
12 4.798744e-04 0.7674159 0.05616991 0.007912941 0.02868665
13 5.234993e-04 0.7673535 0.05423532 0.007886665 0.03045862
14 5.671242e-04 0.7669165 0.04527588 0.007264952 0.03347114
15 6.107492e-04 0.7671033 0.03028188 0.006812843 0.03301975
```

▼ Code

```
plot(tree.cv.alt)
```



▼ Code

```
confusionMatrix(tree.cv.alt$pred$pred, tree.cv.alt$pred$obs)
```

```
Confusion Matrix and Statistics

          Reference
Prediction     Low Borderline  High
  Low        12138       1625  1756
  Borderline    49         34    40
  High         149        105   124

Overall Statistics
```

```
               Accuracy : 0.7675
                 95% CI : (0.7609, 0.7741)
    No Information Rate : 0.77
    P-Value [Acc > NIR] : 0.7767

                  Kappa : 0.0715

 Mcnemar's Test P-Value : <2e-16

Statistics by Class:

                     Class: Low Class: Borderline Class: High
Sensitivity             0.98395          0.019274     0.06458
Specificity             0.08225          0.993757     0.98199
Pos Pred Value          0.78214          0.276423     0.32804
Neg Pred Value          0.60479          0.891174     0.88518
Prevalence              0.77004          0.110112     0.11985
Detection Rate          0.75768          0.002122     0.00774
Detection Prevalence    0.96873          0.007678     0.02360
Balanced Accuracy       0.53310          0.506516     0.52328
```

▼ Code

```
# ALT CV
set.seed(3888)
ctrl = trainControl(method = "repeatedcv", number = 10, repeats
                    savePredictions = "final")
tree.cv.ggt = train(GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKH
                    data = fdat.full, method = "rpart", trContro
                    tuneLength = 15, metric = "Accuracy")
tree.cv.ggt
```

```
CART

5340 samples
   5 predictor
   3 classes: 'Low', 'Borderline', 'High'

No pre-processing
Resampling: Cross-Validated (10 fold, repeated 3 times)
Summary of sample sizes: 4806, 4806, 4805, 4805, 4807, 4805,
...
Resampling results across tuning parameters:

  cp            Accuracy   Kappa
  0.000000e+00  0.8124852  -0.0004799968
```

```
  4.302926e-05  0.8124852  -0.0004799968
  8.605852e-05  0.8124852  -0.0012219830
  1.290878e-04  0.8127967  -0.0011464525
  1.721170e-04  0.8127967  -0.0011464525
  2.151463e-04  0.8127967  -0.0011464525
  2.581756e-04  0.8130469  -0.0007252294
  3.012048e-04  0.8134838   0.0000000000
  3.442341e-04  0.8134838   0.0000000000
  3.872633e-04  0.8134838   0.0000000000
  4.302926e-04  0.8134838   0.0000000000
  4.733219e-04  0.8134838   0.0000000000
  5.163511e-04  0.8134838   0.0000000000
  5.593804e-04  0.8134838   0.0000000000
  6.024096e-04  0.8134838   0.0000000000

Accuracy was used to select the optimal model using the largest
value.
The final value used for the model was cp = 0.0006024096.
```

▼ Code

```
tree.cv.ggt$bestTune
```

```
            cp
15 0.0006024096
```

▼ Code

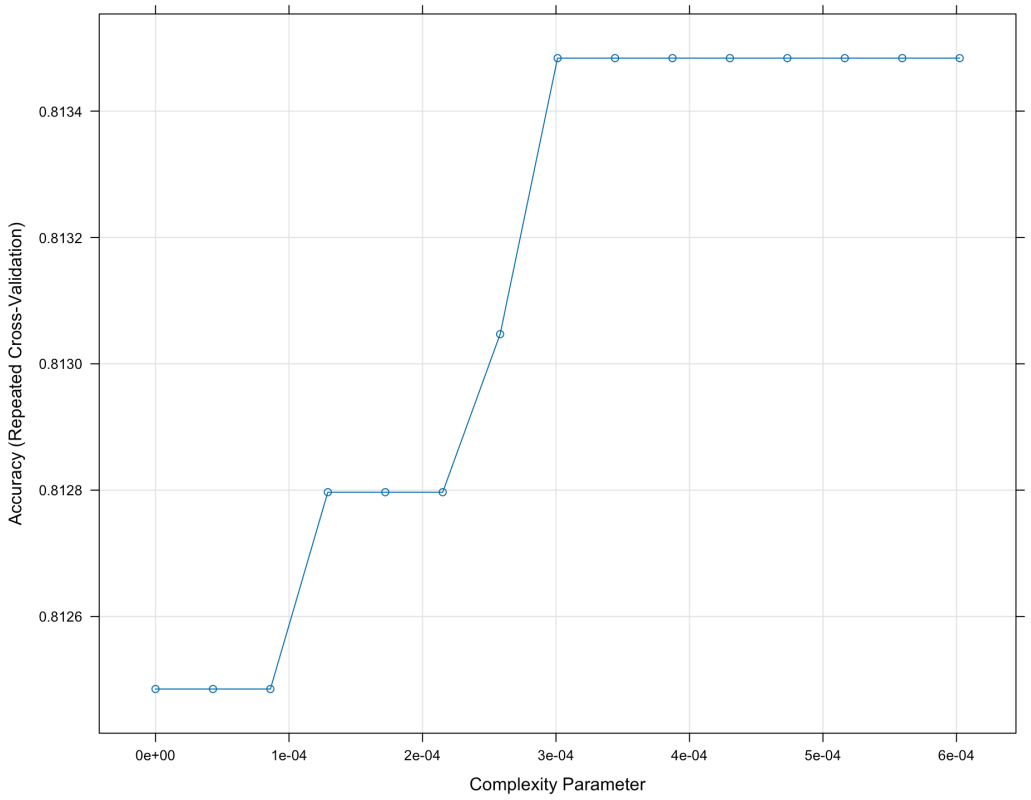
```
tree.cv.ggt$results
```

```
             cp  Accuracy         Kappa   AccuracySD
KappaSD
1  0.000000e+00 0.8124852 -0.0004799968 0.0024643661
0.005420402
2  4.302926e-05 0.8124852 -0.0004799968 0.0024643661
0.005420402
3  8.605852e-05 0.8124852 -0.0012219830 0.0024643661
0.003314895
4  1.290878e-04 0.8127967 -0.0011464525 0.0018294427
0.003316103
5  1.721170e-04 0.8127967 -0.0011464525 0.0018294427
0.003316103
6  2.151463e-04 0.8127967 -0.0011464525 0.0018294427
0.003316103
7  2.581756e-04 0.8130469 -0.0007252294 0.0017515286
0.003009803
8  3.012048e-04 0.8134838  0.0000000000 0.0008795763
```

```
0.000000000
9  3.442341e-04 0.8134838  0.0000000000 0.0008795763
0.000000000
10 3.872633e-04 0.8134838  0.0000000000 0.0008795763
0.000000000
11 4.302926e-04 0.8134838  0.0000000000 0.0008795763
0.000000000
12 4.733219e-04 0.8134838  0.0000000000 0.0008795763
0.000000000
13 5.163511e-04 0.8134838  0.0000000000 0.0008795763
0.000000000
14 5.593804e-04 0.8134838  0.0000000000 0.0008795763
0.000000000
15 6.024096e-04 0.8134838  0.0000000000 0.0008795763
0.000000000
```

▼ Code

```
plot(tree.cv.ggt)
```



▼ Code

```
confusionMatrix(tree.cv.ggt$pred$pred, tree.cv.ggt$pred$obs)
```

```
Confusion Matrix and Statistics

            Reference
Prediction     Low Borderline  High
  Low        13032        882  2106
  Borderline     0          0     0
  High           0          0     0

Overall Statistics

               Accuracy : 0.8135
                 95% CI : (0.8074, 0.8195)
    No Information Rate : 0.8135
    P-Value [Acc > NIR] : 0.5049

                  Kappa : 0

 Mcnemar's Test P-Value : NA

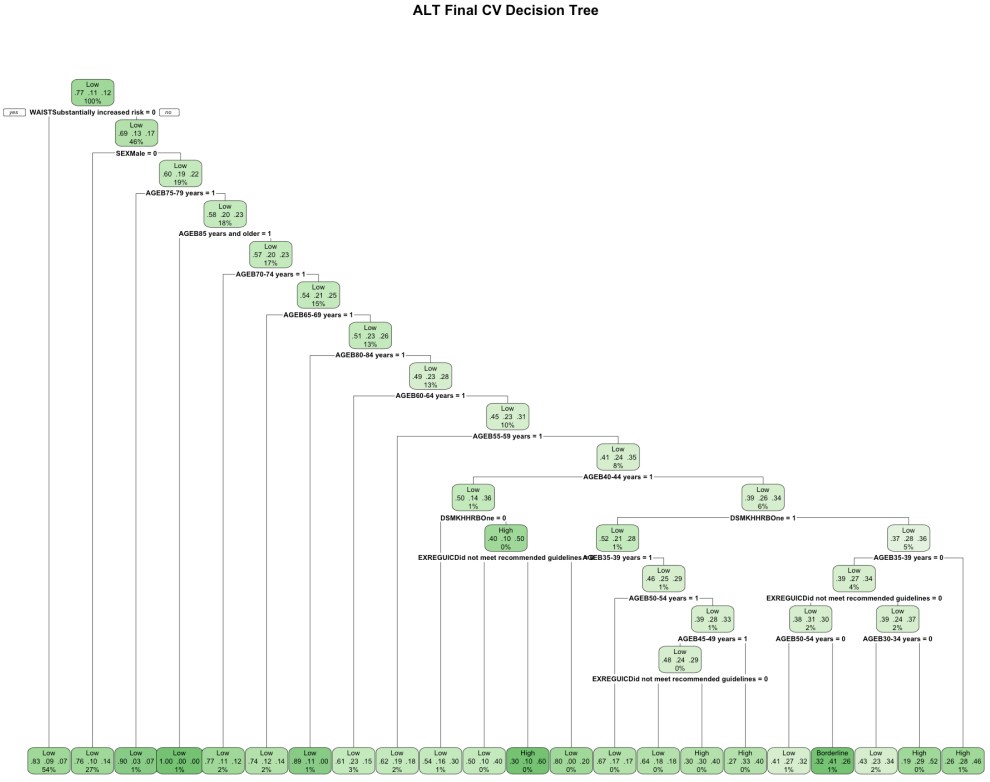
Statistics by Class:

                     Class: Low Class: Borderline Class: High
Sensitivity              1.0000           0.00000      0.0000
Specificity              0.0000           1.00000      1.0000
Pos Pred Value           0.8135               NaN         NaN
Neg Pred Value              NaN           0.94494      0.8685
Prevalence               0.8135           0.05506      0.1315
Detection Rate           0.8135           0.00000      0.0000
Detection Prevalence     1.0000           0.00000      0.0000
Balanced Accuracy        0.5000           0.50000      0.5000
```

However, trying to fit the final trees yields just one node, so I am not really sure if there is something wrong with the code or not. Essentially, it just classified everything as `Low`. So that could explain the respectable accuracy but terrible sensitivity for `Borderline`/`High`, at least for GGT. We also output the variable importance for the original pruned trees as well.

▼ Code

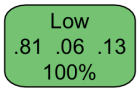
```
# Final CV Trees
rpart.plot(tree.cv.alt$finalModel, box.palette = "Greens",
           type = 2, fallen.leaves = TRUE, tweak = 1.2,
           main = "ALT Final CV Decision Tree")
```



▼ Code

```
rpart.plot(tree.cv.ggt$finalModel, box.palette = "Greens",
           type = 2, fallen.leaves = TRUE, tweak = 1.2,
           main = "GGT Final CV Decision Tree")
```

GGT Final CV Decision Tree



▼ Code

```
# Original pruned trees variable importance
alt.vi = res.alt.pruned$variable.importance %>% as.data.frame()
colnames(alt.vi) = "Mean Decrease in Impurity"
ggt.vi = res.ggt.pruned$variable.importance %>% as.data.frame()
colnames(ggt.vi) = "Mean Decrease in Impurity"
kable(alt.vi, caption = "ALT Tree", digits = 3, align = "c")
```

ALT Tree

	Mean Decrease in Impurity
AGEB	54.412
WAIST	40.797
SEX	31.017
EXREGUIC	4.798
DSMKHHRB	3.593

▼ Code

```
kable(ggt.vi, caption = "GGT Tree", digits = 3, align = "c")
```

GGT Tree

	Mean Decrease in Impurity
WAIST	39.516
AGEB	13.252
EXREGUIC	5.139
DSMKHHRB	3.800
SEX	1.863

Random Forests

We first start by fitting the random forest using the default `mtry`, which is 2 in our case.

▼ Code

```
set.seed(3858)
# ALT
alt.rf = randomForest(ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSM
                      data = fdat.full,
                      importance = TRUE)
alt.rf
```

```
Call:
 randomForest(formula = ALT3 ~ EXREGUIC + AGEB + SEX + WAIST +
DSMKHHRB, data = fdat.full, importance = TRUE)
               Type of random forest: classification
                     Number of trees: 500
No. of variables tried at each split: 2

        OOB estimate of  error rate: 23.01%
Confusion matrix:
           Low Borderline High  class.error
Low        4111          0    1 0.0002431907
Borderline  588          0    0 1.0000000000
High        640          0    0 1.0000000000
```

▼ Code

```
alt.class.error = 1 - confusionMatrix(fdat.full$ALT3,
                                      predict(alt.rf, newdata =
```

```
                                                    type = "class"))$
alt.class.error
```

```
       Class: Low Class: Borderline       Class: High
        0.2295297         0.0000000         0.0000000
```

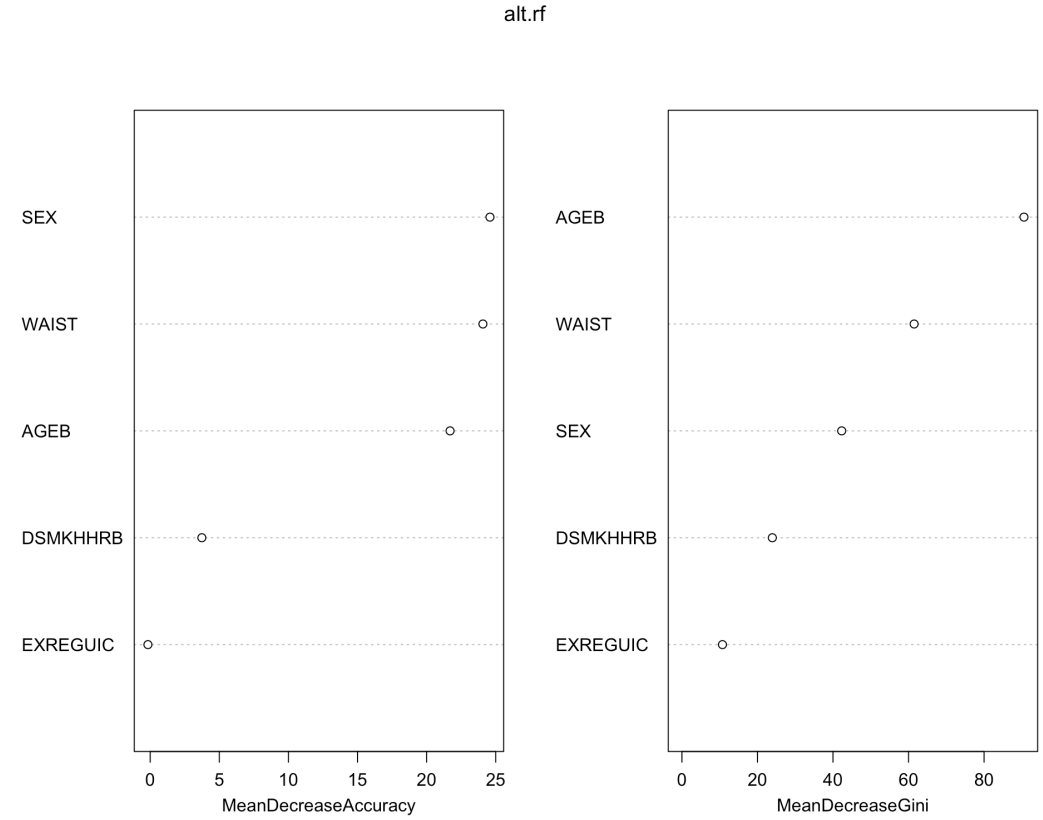
▼ Code

```
alt.rf$importance
```

```
                  Low   Borderline         High
MeanDecreaseAccuracy
EXREGUIC 0.0001803982 -0.001851905 3.545353e-04
-1.959286e-05
AGEB     0.0083796303 -0.001563674 4.433760e-03
6.806091e-03
SEX      0.0078867627  0.003245900 9.004668e-03
7.501287e-03
WAIST    0.0106029001 -0.003027665 5.790892e-03
8.528196e-03
DSMKHHRB 0.0007995909 -0.001133185 1.607503e-05
4.926085e-04
         MeanDecreaseGini
EXREGUIC         10.73526
AGEB             90.56051
SEX              42.29963
WAIST            61.48091
DSMKHHRB         23.93749
```

▼ Code

```
varImpPlot(alt.rf)
```



▼ Code

```
# GGT
ggt.rf = randomForest(GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSM
                      data = fdat.full,
                      importance = TRUE)
ggt.rf
```

```
Call:
 randomForest(formula = GGT3 ~ EXREGUIC + AGEB + SEX + WAIST +
DSMKHHRB, data = fdat.full, importance = TRUE)
               Type of random forest: classification
                     Number of trees: 500
No. of variables tried at each split: 2

        OOB estimate of  error rate: 18.63%
Confusion matrix:
           Low Borderline High class.error
Low        4344          0    0   0.0000000
Borderline  294          0    0   1.0000000
High        701          0    1   0.9985755
```

▼ Code

```
ggt.class.error = 1 - confusionMatrix(fdat.full$GGT3,
                                      predict(ggt.rf, newdata =
                                              type = "class"))$
ggt.class.error
```

```
      Class: Low Class: Borderline       Class: High
       0.1862121                NA         0.0000000
```

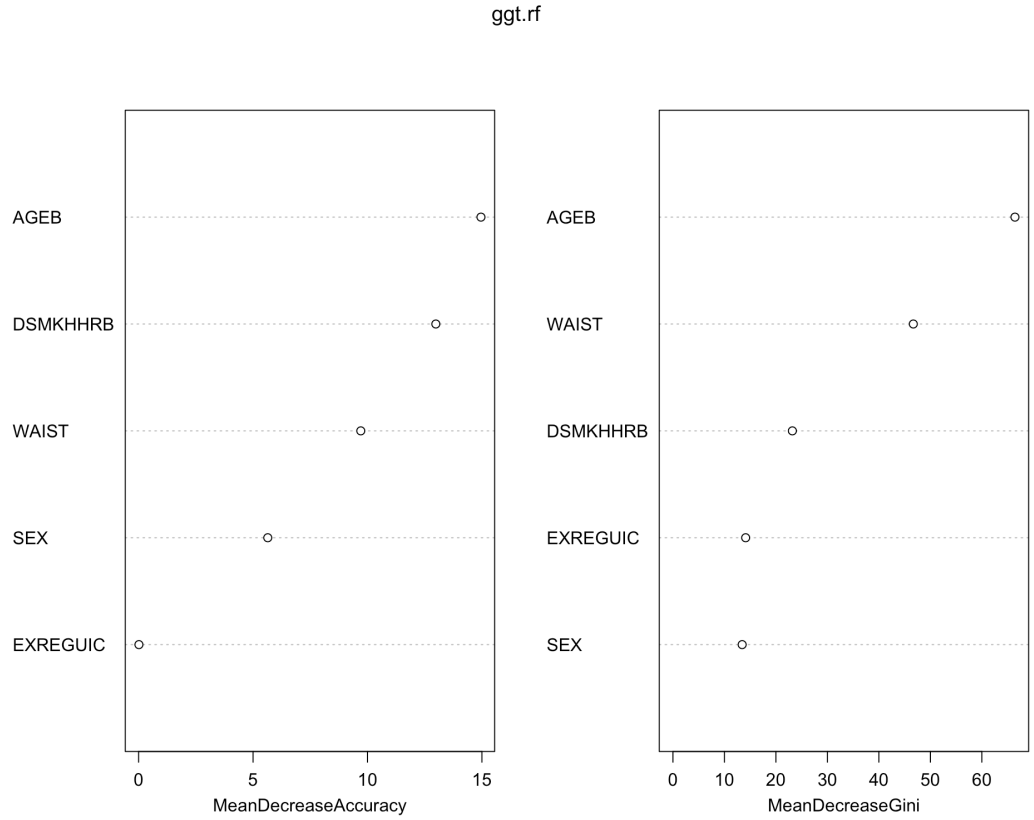
▼ Code

```
ggt.rf$importance
```

```
                    Low    Borderline          High
MeanDecreaseAccuracy
EXREGUIC -0.0002028771  0.0011097394  7.758871e-04
1.400807e-06
AGEB      0.0032042988 -0.0002864713 -2.123824e-03
2.310935e-03
SEX       0.0008202731  0.0009635123 -1.243804e-03
5.573872e-04
WAIST     0.0019199899  0.0004359512 -2.268056e-03
1.286305e-03
DSMKHHRB  0.0021704659  0.0016737387  6.814258e-05
1.868291e-03
         MeanDecreaseGini
EXREGUIC         14.12429
AGEB             66.42685
SEX              13.44743
WAIST            46.68487
DSMKHHRB         23.19777
```

▼ Code

```
varImpPlot(ggt.rf)
```



Next, we do cross-validation to determine the best `mtry`.

▼ Code

```
set.seed(3888)

# ALT
ctrl = trainControl(method = "cv", number = 10, savePredictions
if (file.exists(here("Objects", "alt_rf_train.rds"))) {

  alt.cv.rf = readRDS(here("Objects", "alt_rf_train.rds"))

} else {

  alt.cv.rf = train(ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMK
                    data = fdat.full, method = "rf", trControl =
                    tuneGrid = expand.grid(mtry = 1:5), ntree = 5
                    importance = TRUE, metric = "Accuracy")
  saveRDS(alt.cv.rf, here("Objects", "alt_rf_train.rds"))

}
# CV tuning results
alt.cv.rf$results
```

```
  mtry  Accuracy       Kappa   AccuracySD     KappaSD
1    1 0.7700375 0.000000000 0.0007895824 0.000000000
2    2 0.7700375 0.000000000 0.0007895824 0.000000000
3    3 0.7700375 0.000000000 0.0007895824 0.000000000
4    4 0.7702247 0.004451623 0.0012639486 0.008863873
5    5 0.7713483 0.027166783 0.0027137410 0.016663666
```

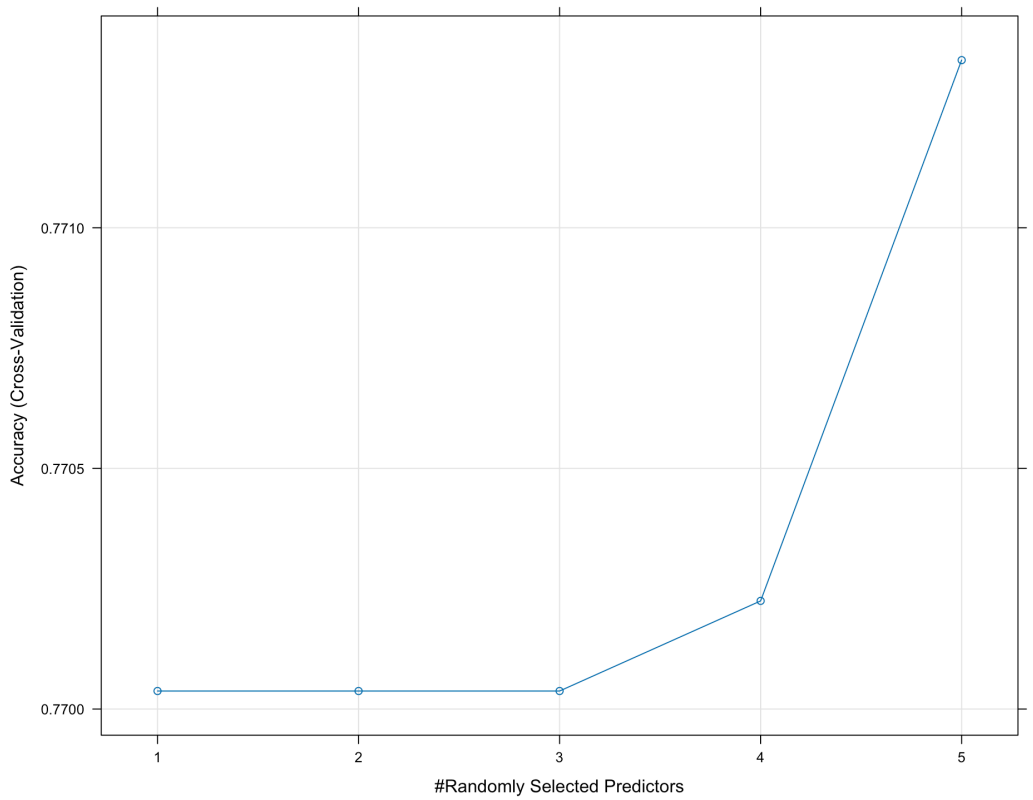
▼ Code

```
alt.cv.rf$bestTune
```

```
  mtry
5    5
```

▼ Code

```
plot(alt.cv.rf)
```



▼ Code

```
confusionMatrix(alt.cv.rf$pred$pred, alt.cv.rf$pred$obs)
```

```
Confusion Matrix and Statistics
```

```
          Reference
Prediction    Low Borderline High
  Low        4101        575  622
  Borderline    0          0    0
  High         11         13   18

Overall Statistics

               Accuracy : 0.7713
                 95% CI : (0.7598, 0.7826)
    No Information Rate : 0.77
    P-Value [Acc > NIR] : 0.4174

                  Kappa : 0.0273

 Mcnemar's Test P-Value : <2e-16

Statistics by Class:

                     Class: Low Class: Borderline Class: High
Sensitivity             0.99732            0.0000    0.028125
Specificity             0.02524            1.0000    0.994894
Pos Pred Value          0.77407               NaN    0.428571
Neg Pred Value          0.73810            0.8899    0.882597
Prevalence              0.77004            0.1101    0.119850
Detection Rate          0.76798            0.0000    0.003371
Detection Prevalence    0.99213            0.0000    0.007865
Balanced Accuracy       0.51128            0.5000    0.511509
```

▼ Code

```
varImp(alt.cv.rf)
```

```
rf variable importance

  variables are sorted by maximum importance across the classes
  only 20 most important variables shown (out of 21)

                                             Low Borderline
High
SEXMale                                  100.000    37.2317
54.579
WAISTSubstantially increased risk         58.727     0.0000
37.025
WAISTNot at risk                          53.209     6.4389
32.279
AGEB35-39 years                           29.351    13.8027
```

```
41.692
AGEB30-34 years                                 31.423    16.2904
36.838
AGEB80-84 years                                 19.500    21.2527
34.665
AGEB20-24 years                                 32.560     6.8696
8.765
AGEB75-79 years                                 25.478    16.6079
31.253
AGEB50-54 years                                  9.513    27.7482
18.105
AGEB55-59 years                                 26.937     5.9751
2.619
AGEB85 years and older                          25.020    16.6223
26.821
AGEB65-69 years                                 25.744    14.0553
12.571
AGEB70-74 years                                 24.482    14.6146
16.091
AGEB60-64 years                                 23.314    10.8412
4.930
AGEB40-44 years                                 12.265    11.4298
23.165
DSMKHHRBThree or more                            7.668    22.5468
5.514
AGEB45-49 years                                 12.115    12.8750
21.953
EXREGUICDid not meet recommended guidelines     19.398     5.8655
14.127
DSMKHHRBOne                                     14.234     2.2984
17.900
DSMKHHRBTwo                                     12.758     0.3265
9.618
```

▼ Code

```
# GGT
set.seed(3888)
ctrl = trainControl(method = "cv", number = 10, savePredictions
if (file.exists(here("Objects", "ggt_rf_train.rds"))) {

  ggt.cv.rf = readRDS(here("Objects", "ggt_rf_train.rds"))

} else {

  ggt.cv.rf = train(GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKH
                    data = fdat.full, method = "rf", trControl = 
                    tuneGrid = expand.grid(mtry = 1:5), ntree = 5(
```

```
                    importance = TRUE, metric = "Accuracy")
  saveRDS(ggt.cv.rf, here("Objects", "ggt_rf_train.rds"))

}
# CV tuning results
ggt.cv.rf$results
```

```
  mtry Accuracy Kappa   AccuracySD KappaSD
1    1 0.813484     0 0.0006803877       0
2    2 0.813484     0 0.0006803877       0
3    3 0.813484     0 0.0006803877       0
4    4 0.813484     0 0.0006803877       0
5    5 0.813484     0 0.0006803877       0
```

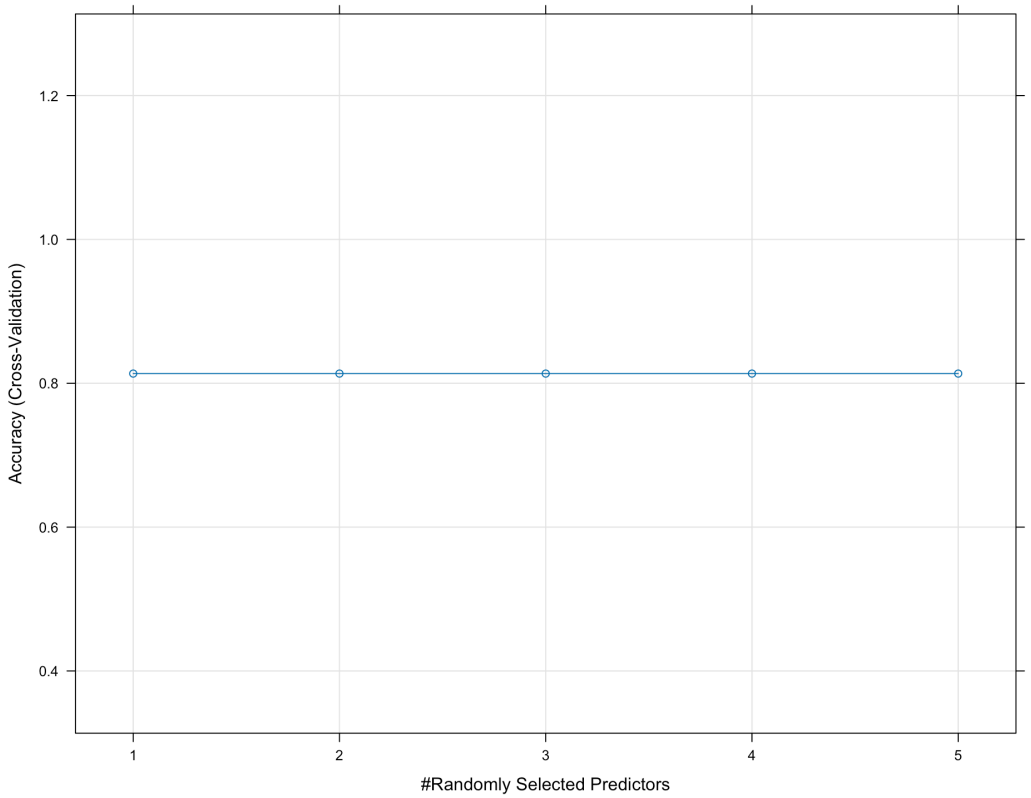
▼ Code

```
ggt.cv.rf$bestTune
```

```
  mtry
1    1
```

▼ Code

```
plot(ggt.cv.rf)
```



▼ Code

```
confusionMatrix(ggt.cv.rf$pred$pred, ggt.cv.rf$pred$obs)
```

```
Confusion Matrix and Statistics

           Reference
Prediction    Low Borderline High
  Low        4344        294  702
  Borderline    0          0    0
  High          0          0    0

Overall Statistics

               Accuracy : 0.8135
                 95% CI : (0.8028, 0.8239)
    No Information Rate : 0.8135
    P-Value [Acc > NIR] : 0.5085

                  Kappa : 0

 Mcnemar's Test P-Value : NA

Statistics by Class:
```

	Class: Low	Class: Borderline	Class: High
Sensitivity	1.0000	0.00000	0.0000
Specificity	0.0000	1.00000	1.0000
Pos Pred Value	0.8135	NaN	NaN
Neg Pred Value	NaN	0.94494	0.8685
Prevalence	0.8135	0.05506	0.1315
Detection Rate	0.8135	0.00000	0.0000
Detection Prevalence	1.0000	0.00000	0.0000
Balanced Accuracy	0.5000	0.50000	0.5000

▼ Code

```
varImp(ggt.cv.rf)
```

```
rf variable importance

  variables are sorted by maximum importance across the classes
  only 20 most important variables shown (out of 21)

                                  Low Borderline   High
AGEB65–69 years                100.00     37.262 92.679
DSMKHHRBTwo                     89.52     37.262 42.414
DSMKHHRBOne                     81.15     37.981 32.192
AGEB80–84 years                 80.31      5.701 36.290
AGEB55–59 years                 41.21     37.262 75.945
AGEB40–44 years                 26.28     37.262 74.391
AGEB25–29 years                 44.06     37.262 68.779
SEXMale                         65.68      5.712 15.533
AGEB20–24 years                 21.87     37.262 59.557
AGEB70–74 years                 56.62     14.967 14.967
AGEB60–64 years                 56.62     37.262 51.934
AGEB85 years and older          54.83     37.262  5.706
AGEB45–49 years                 21.52     37.262 51.202
AGEB75–79 years                 48.51     37.262 50.046
AGEB30–34 years                 47.08     14.967 14.967
WAISTNot at risk                44.18     37.262  2.098
WAISTSubstantially increased risk  43.41     14.967  7.769
DSMKHHRBThree or more           40.49     37.262  7.377
AGEB50–54 years                  0.00     37.262 14.967
AGEB35–39 years                 26.02     14.967 37.262
```

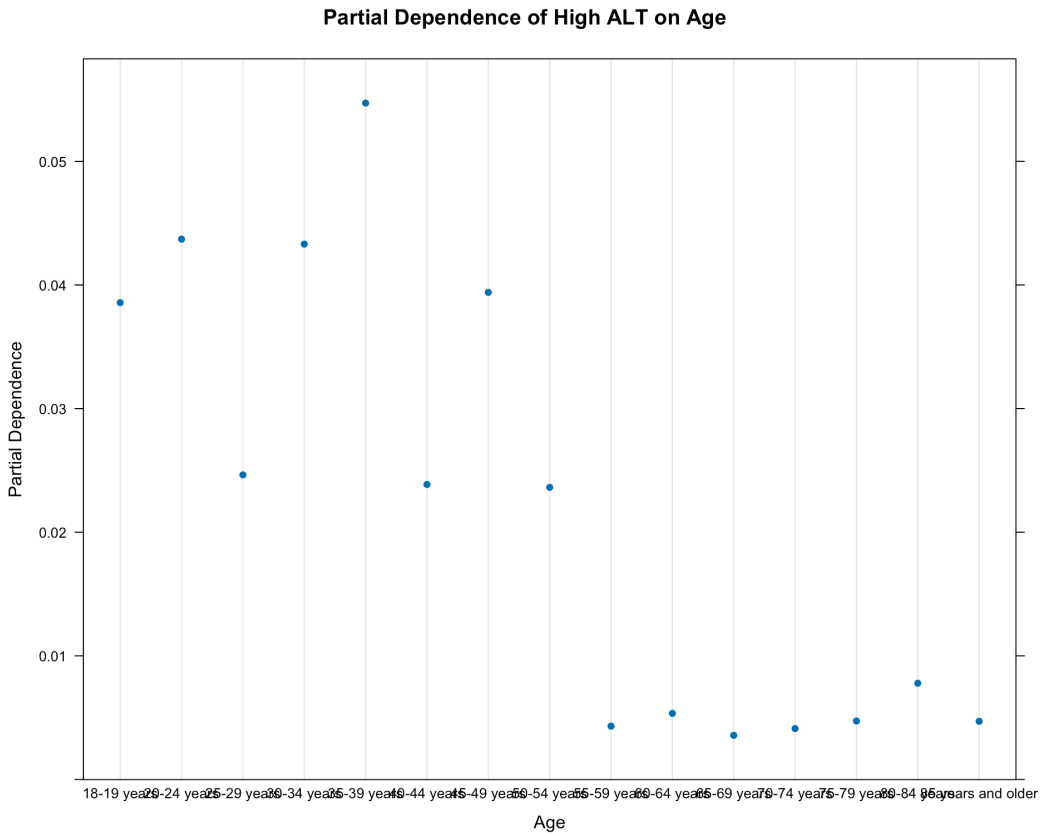
The sensitivities overall say the same story as with decision trees, as it seems `Low` is heavily preferred in both cases especially for GGT.

Partial Dependence Plots

We see the partial plots against `High` for each variable for now (a quick analysis showed that `Borderline` plots are roughly similar, but for brevity I will leave them off). We can expand to more higher-dimensional partial plots later on but for a quick analysis this should be enough to determine if there are nonlinear relationships at play here…

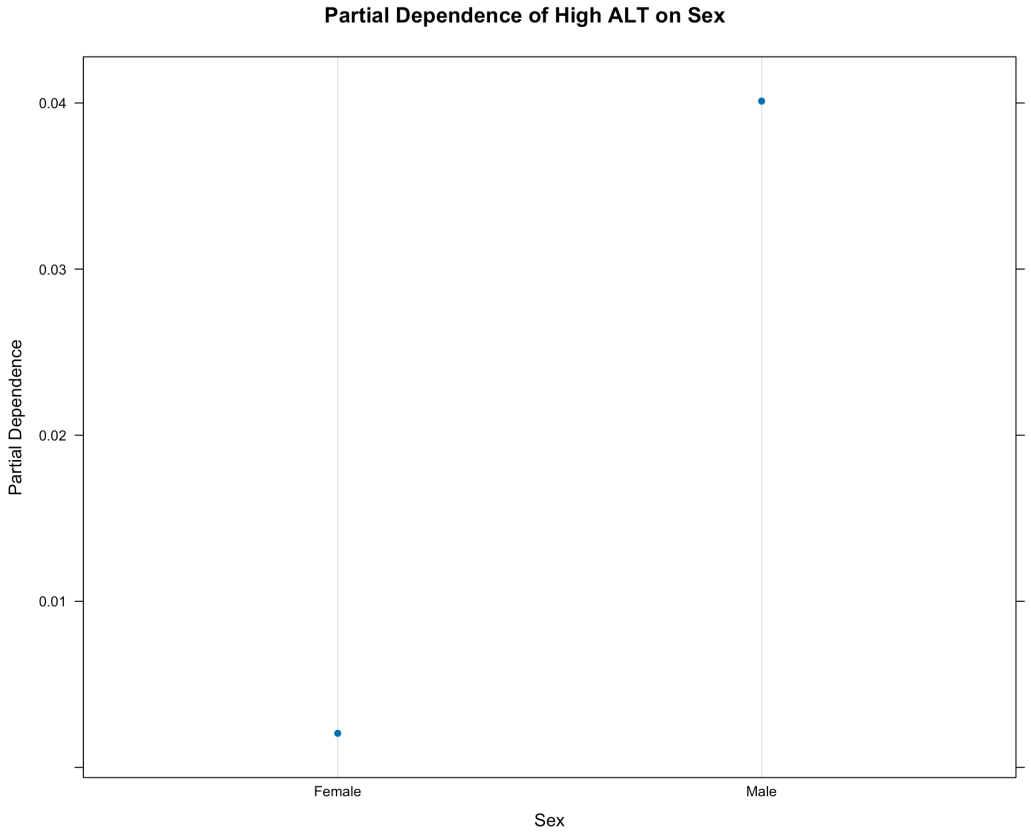
▼ Code

```
# Using default RF mtry = 2; do for each variable
pdp.age.alt = partial(alt.rf, pred.var = "AGEB", train = fdat.fu
                      which.class = "High", prob = TRUE)
plotPartial(pdp.age.alt,
            main = "Partial Dependence of High ALT on Age", xlab
            ylab = "Partial Dependence")
```



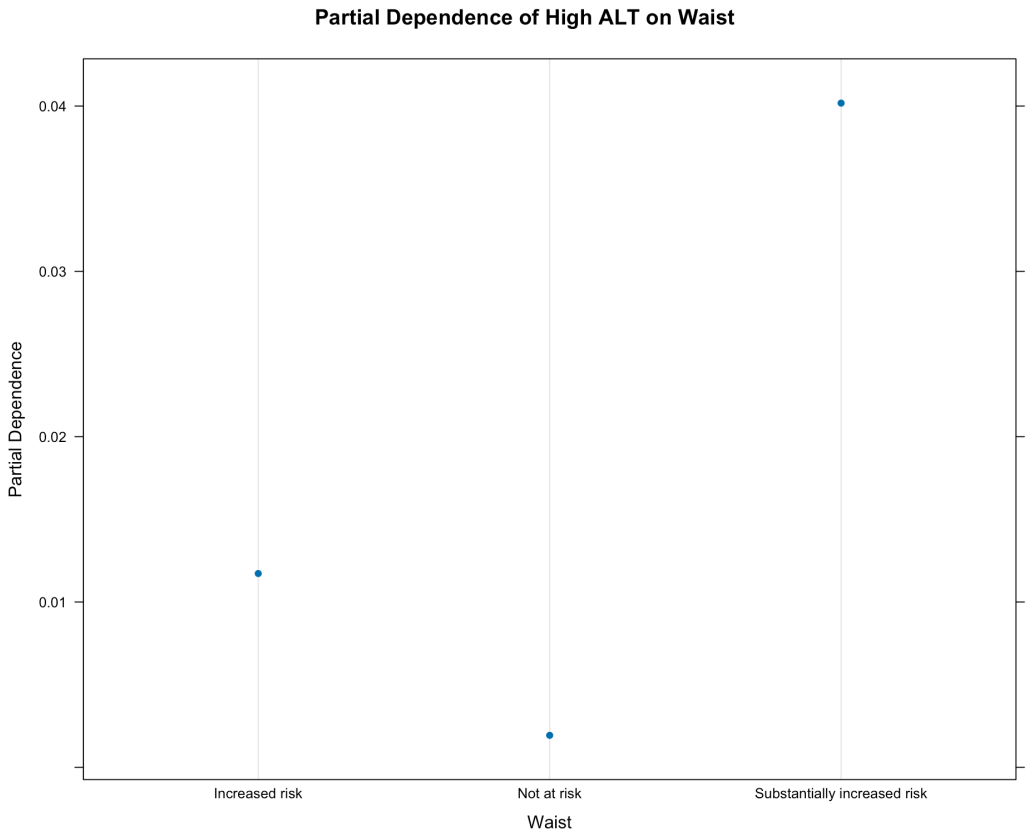
▼ Code

```
pdp.sex.alt = partial(alt.rf, pred.var = "SEX", train = fdat.fu
                      which.class = "High", prob = TRUE)
plotPartial(pdp.sex.alt,
            main = "Partial Dependence of High ALT on Sex", xlab
            ylab = "Partial Dependence")
```



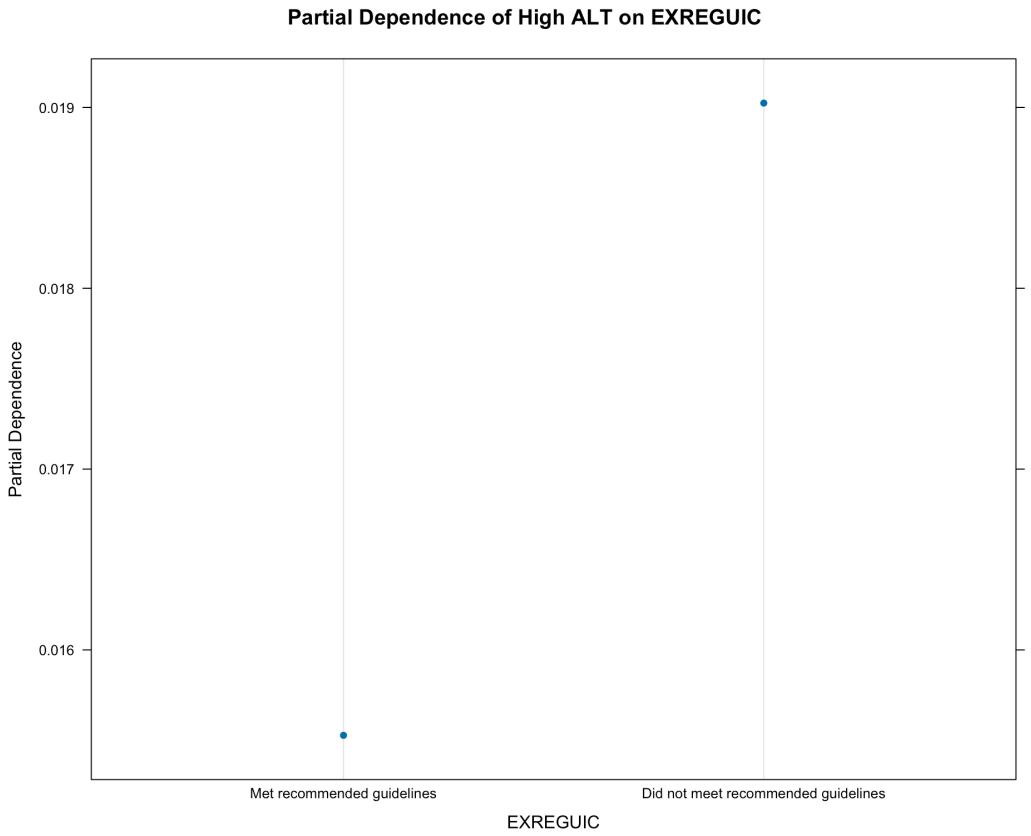
▼ Code

```
pdp.waist.alt = partial(alt.rf, pred.var = "WAIST", train = fda
                        which.class = "High", prob = TRUE)
plotPartial(pdp.waist.alt,
            main = "Partial Dependence of High ALT on Waist", x
            ylab = "Partial Dependence")
```



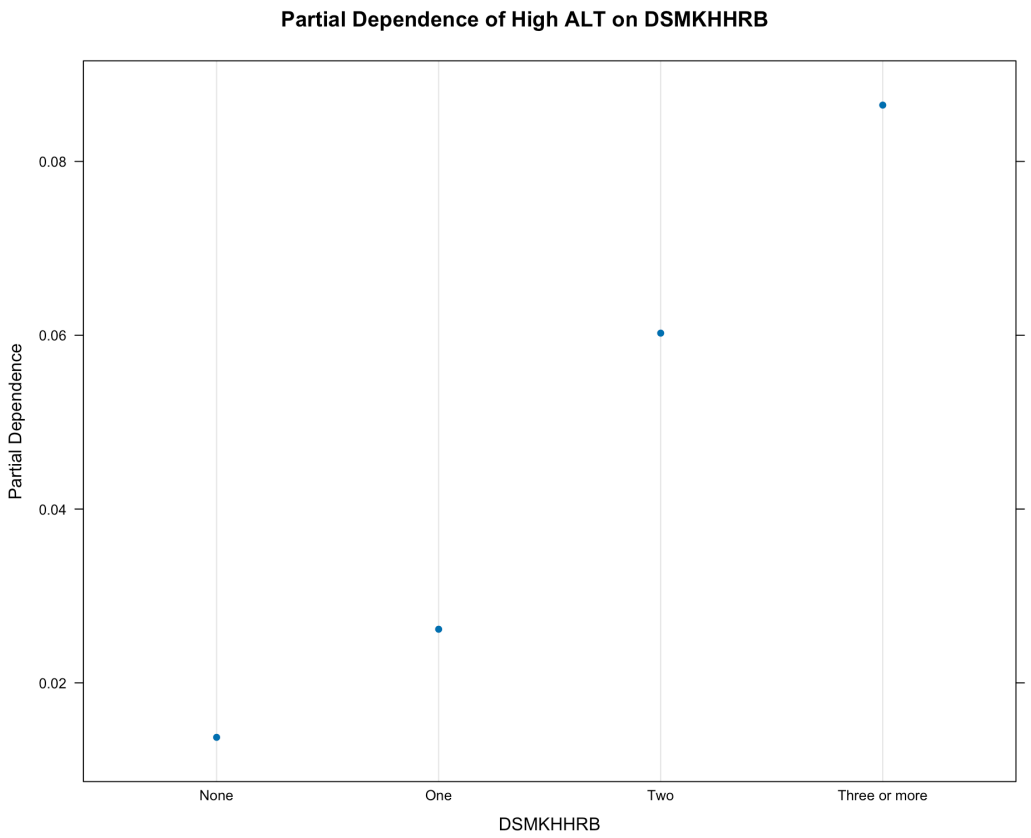
▼ Code

```
pdp.exercise.alt = partial(alt.rf, pred.var = "EXREGUIC", train
                           which.class = "High", prob = TRUE)
plotPartial(pdp.exercise.alt,
            main = "Partial Dependence of High ALT on EXREGUIC",
            ylab = "Partial Dependence")
```



▼ Code

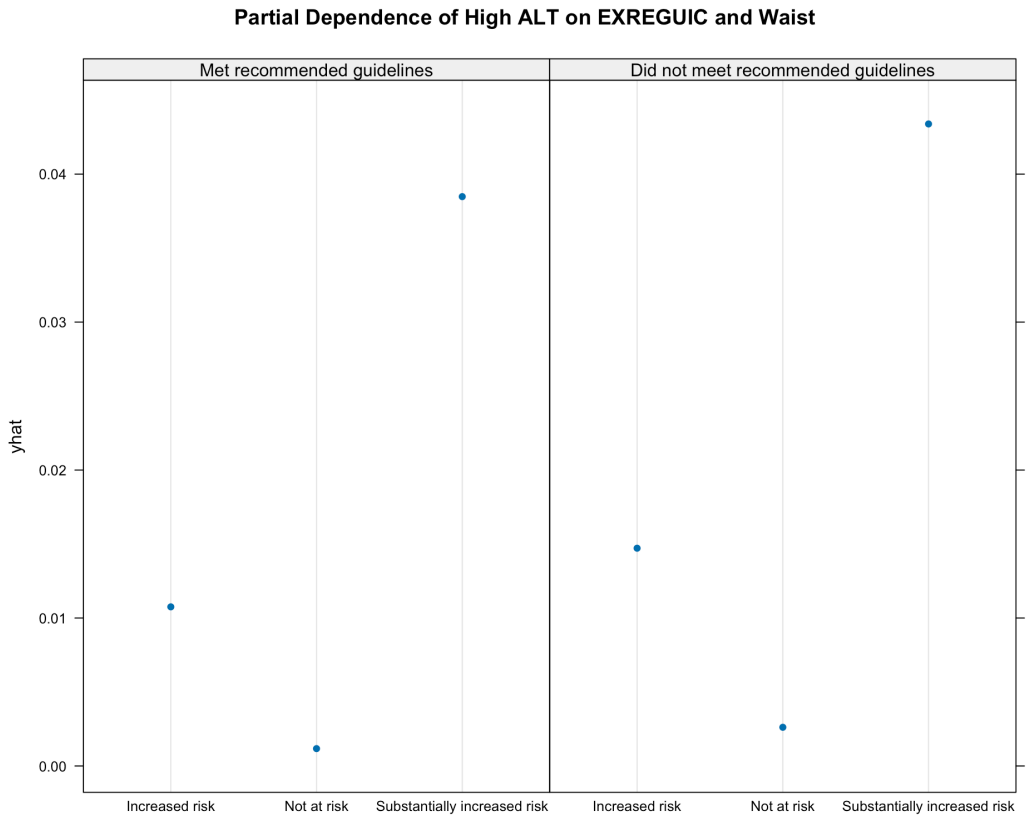
```
pdp.smoking.alt = partial(alt.rf, pred.var = "DSMKHHRB", train =
                          which.class = "High", prob = TRUE)
plotPartial(pdp.smoking.alt,
            main = "Partial Dependence of High ALT on DSMKHHRB",
            ylab = "Partial Dependence")
```



Perhaps, we can do it with `EXREGUIC` and one of the variables, each time.

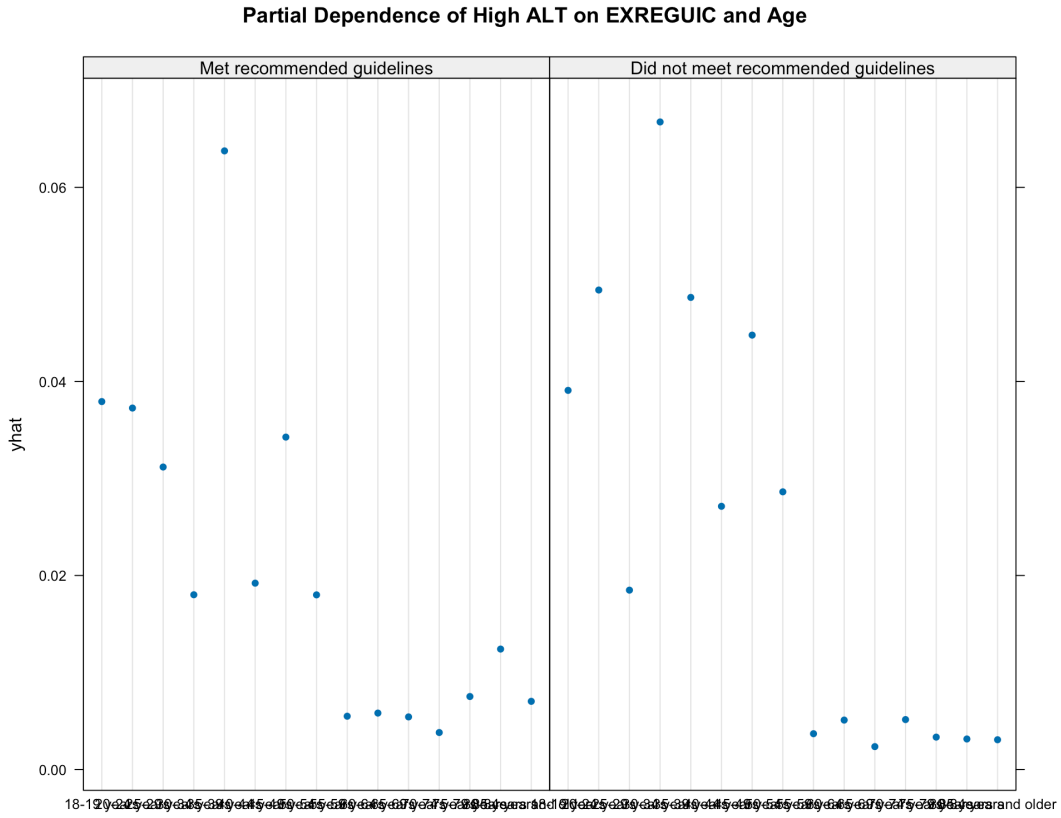
▼ Code

```
# Contour plots
pdp.ex.waist.alt = partial(alt.rf, pred.var = c("WAIST", "EXREGU
                           train = fdat.full, which.class = "Hig
plotPartial(pdp.ex.waist.alt, contour = TRUE,
            main = "Partial Dependence of High ALT on EXREGUIC a
```



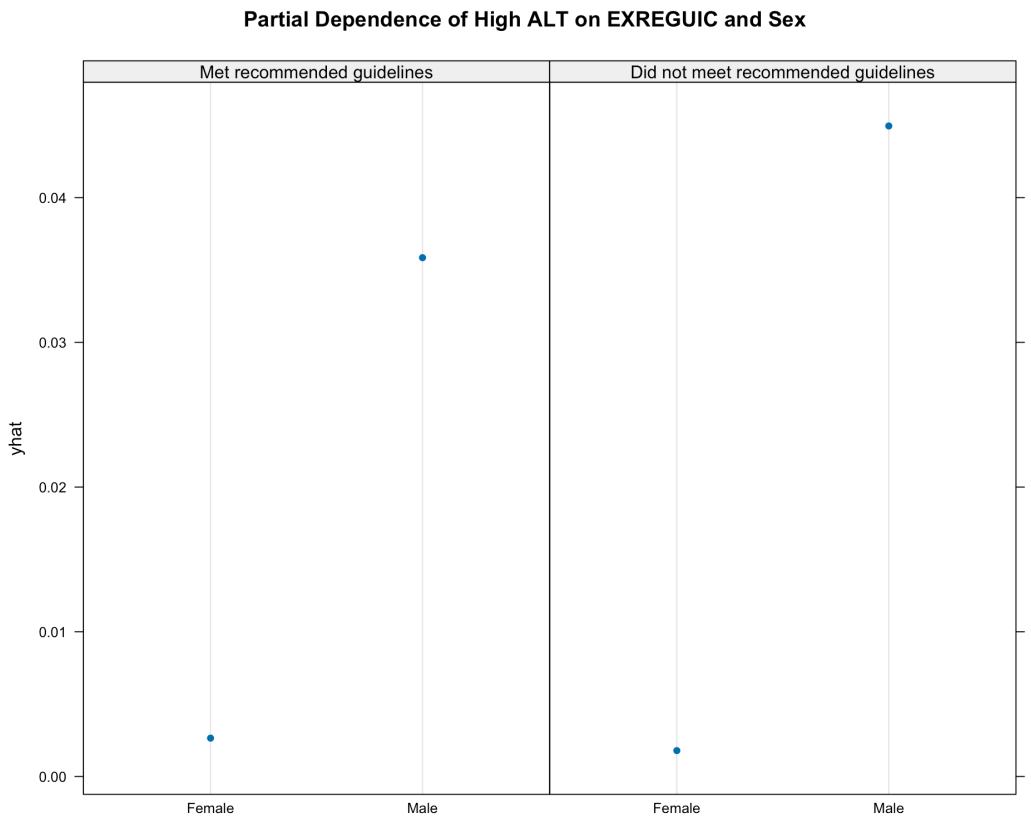
▼ Code

```
pdp.ex.age.alt = partial(alt.rf, pred.var = c("AGEB", "EXREGUIC'
                         train = fdat.full, which.class = "Hig
plotPartial(pdp.ex.age.alt, contour = TRUE,
            main = "Partial Dependence of High ALT on EXREGUIC a
```



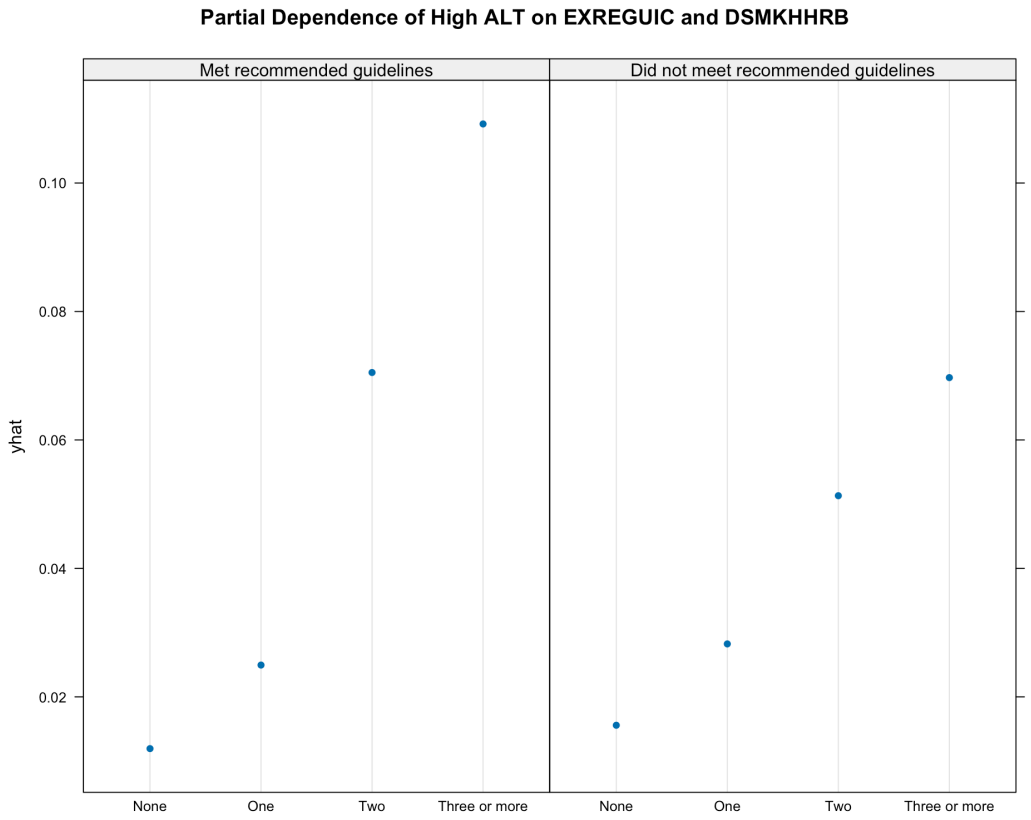
▼ Code

```
pdp.ex.sex.alt = partial(alt.rf, pred.var = c("SEX", "EXREGUIC"
                              train = fdat.full, which.class = "Hig
plotPartial(pdp.ex.sex.alt, contour = TRUE,
            main = "Partial Dependence of High ALT on EXREGUIC a
```



▼ Code

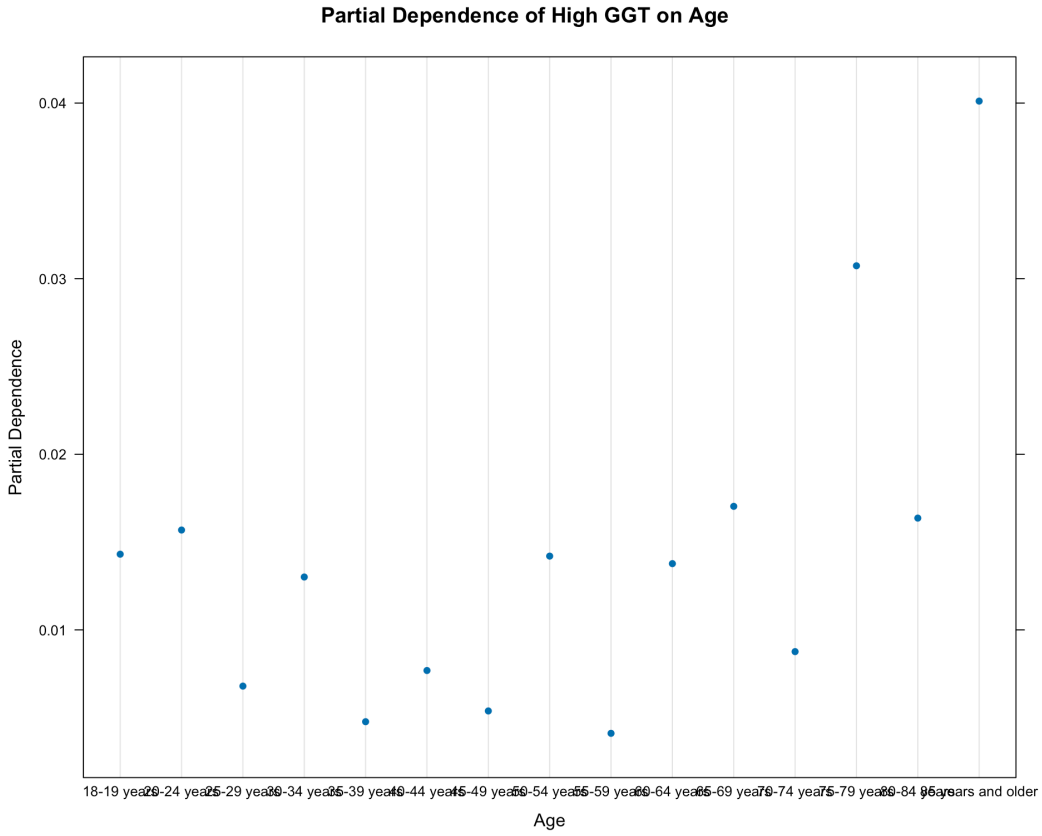
```
pdp.ex.smoking.alt = partial(alt.rf, pred.var = c("DSMKHHRB", "
                             train = fdat.full, which.class = "Hi
plotPartial(pdp.ex.smoking.alt, contour = TRUE,
            main = "Partial Dependence of High ALT on EXREGUIC a
```



Finally, we do the same for GGT as well.

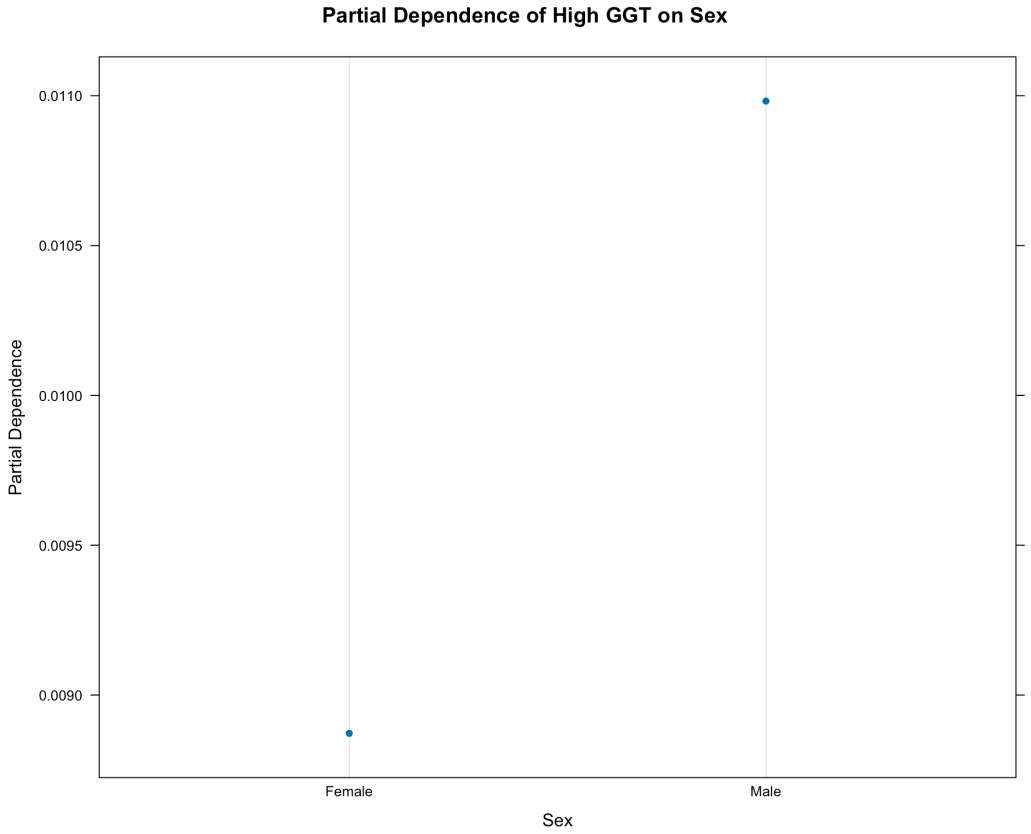
▼ Code

```
# Using default RF mtry = 2; do for each variable
pdp.age.ggt = partial(ggt.rf, pred.var = "AGEB", train = fdat.fu
                      which.class = "High", prob = TRUE)
plotPartial(pdp.age.ggt,
            main = "Partial Dependence of High GGT on Age", xlab
            ylab = "Partial Dependence")
```



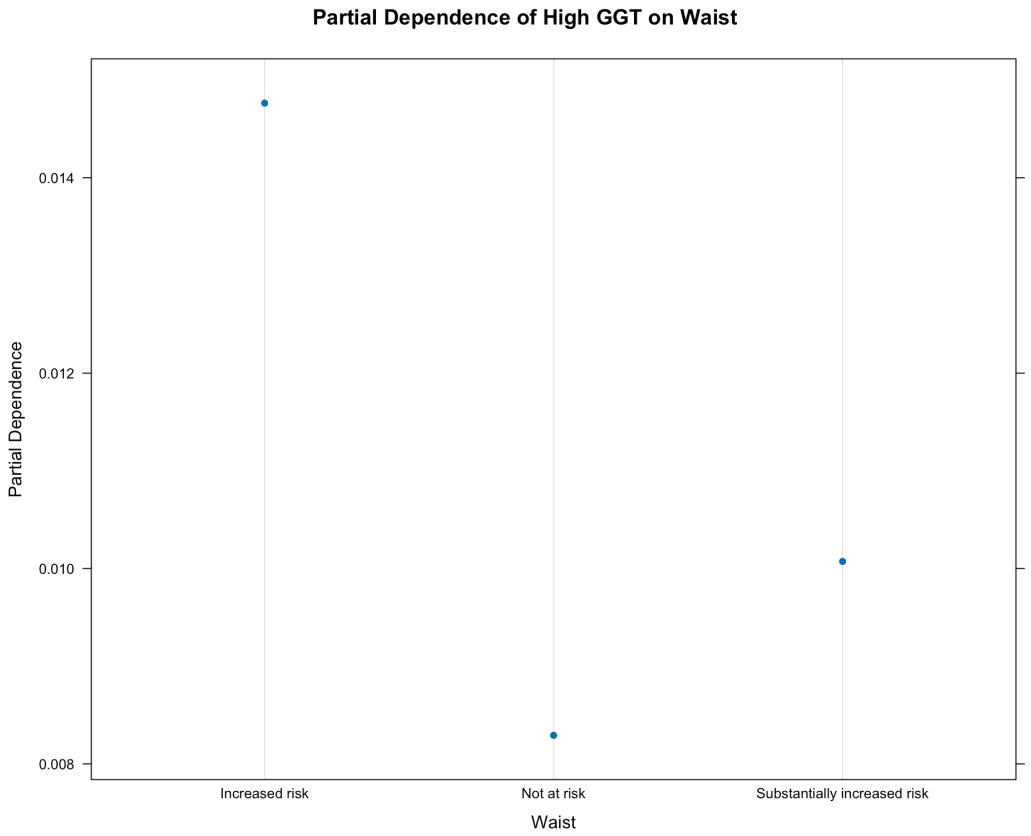
▼ Code

```
pdp.sex.ggt = partial(ggt.rf, pred.var = "SEX", train = fdat.fu
                      which.class = "High", prob = TRUE)
plotPartial(pdp.sex.ggt,
            main = "Partial Dependence of High GGT on Sex", xla
            ylab = "Partial Dependence")
```



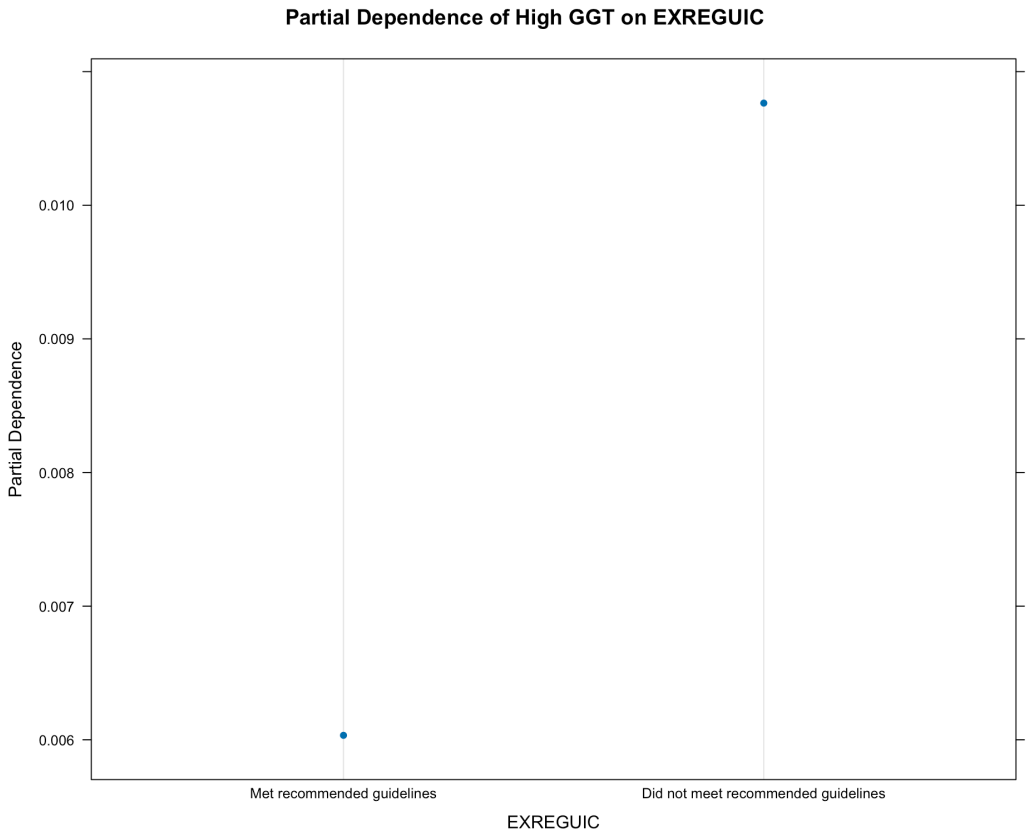
▼ Code

```
pdp.waist.ggt = partial(ggt.rf, pred.var = "WAIST", train = fda
                        which.class = "High", prob = TRUE)
plotPartial(pdp.waist.ggt,
            main = "Partial Dependence of High GGT on Waist", x
            ylab = "Partial Dependence")
```



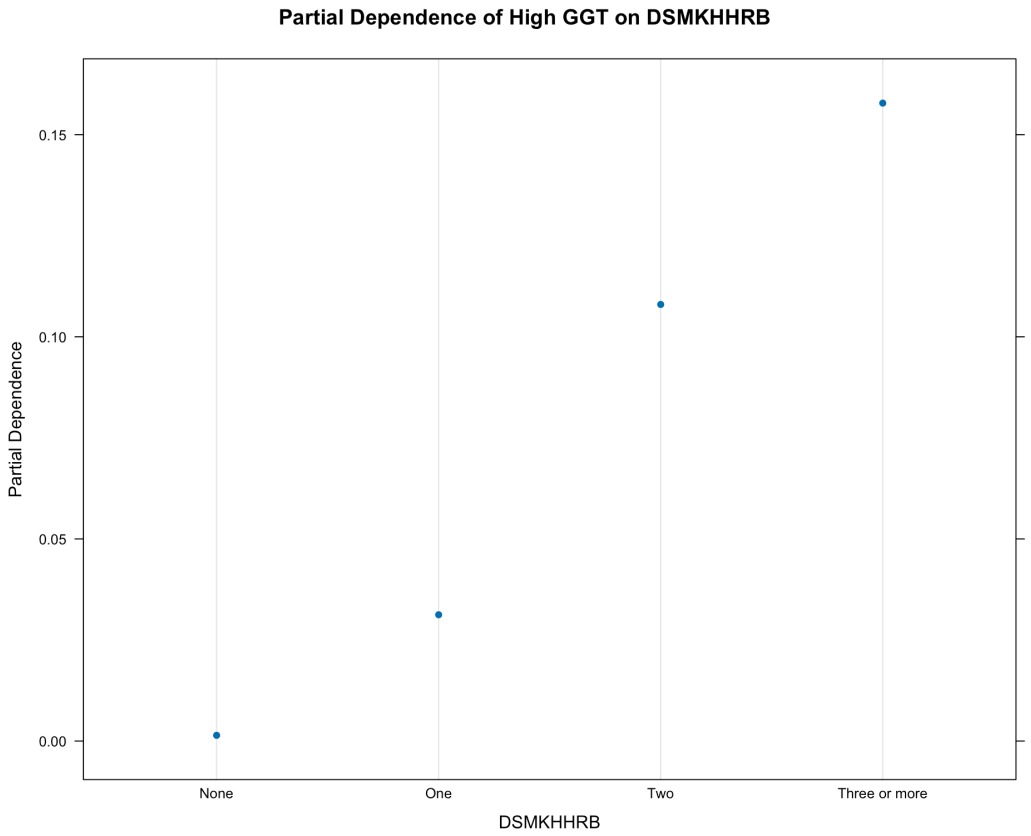
▼ Code

```
pdp.exercise.ggt = partial(ggt.rf, pred.var = "EXREGUIC", train
                      which.class = "High", prob = TRUE)
plotPartial(pdp.exercise.ggt,
            main = "Partial Dependence of High GGT on EXREGUIC",
            ylab = "Partial Dependence")
```



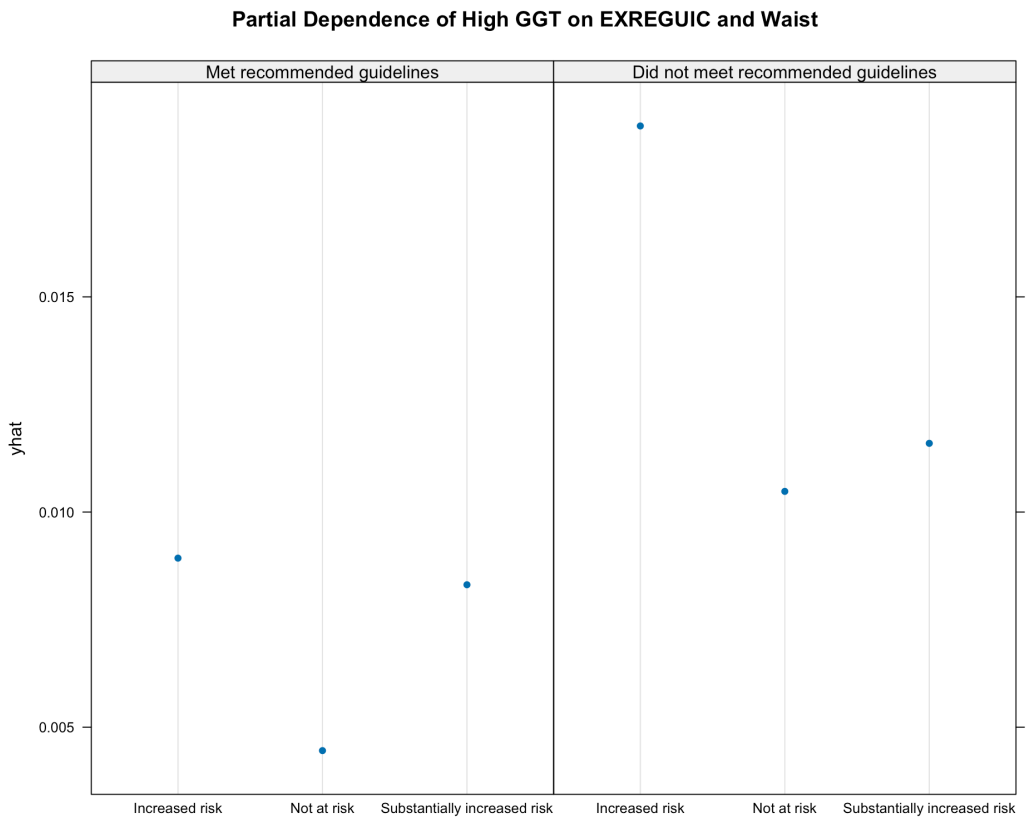
▼ Code

```
pdp.smoking.ggt = partial(ggt.rf, pred.var = "DSMKHHRB", train =
                          which.class = "High", prob = TRUE)
plotPartial(pdp.smoking.ggt,
            main = "Partial Dependence of High GGT on DSMKHHRB",
            ylab = "Partial Dependence")
```



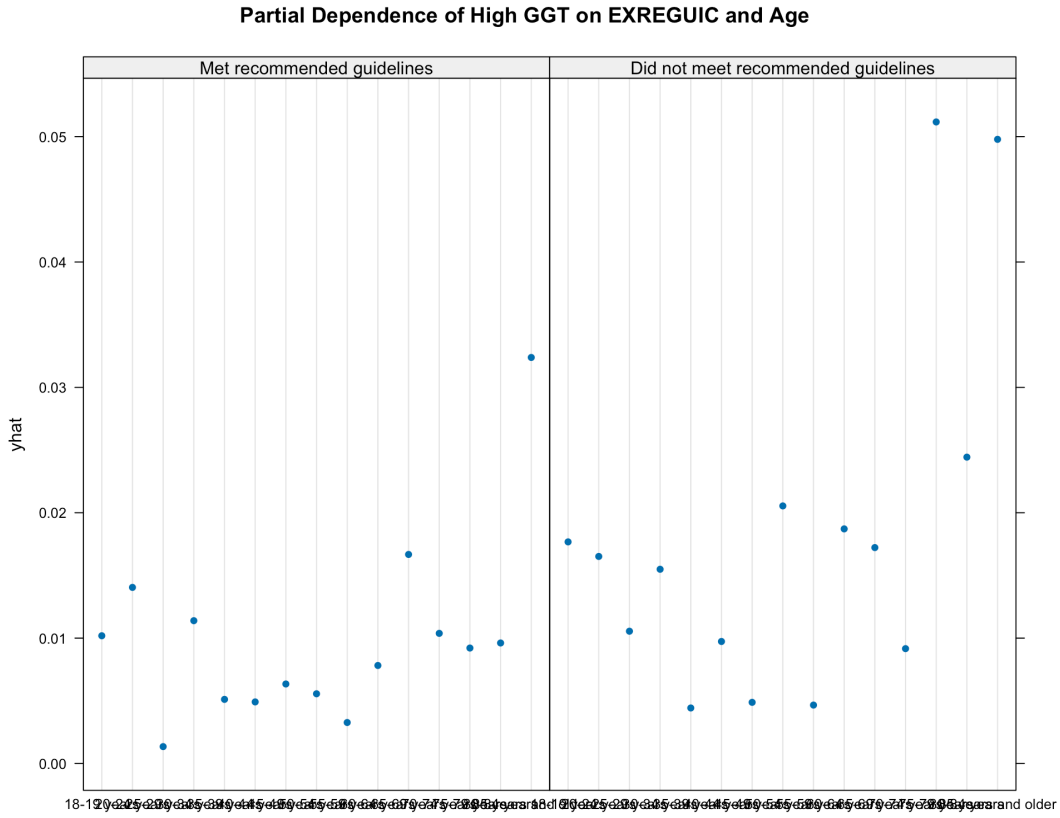
▼ Code

```
# Contour plots
pdp.ex.waist.ggt = partial(ggt.rf, pred.var = c("WAIST", "EXREGU
                           train = fdat.full, which.class = "Hig
plotPartial(pdp.ex.waist.ggt, contour = TRUE,
            main = "Partial Dependence of High GGT on EXREGUIC a
```



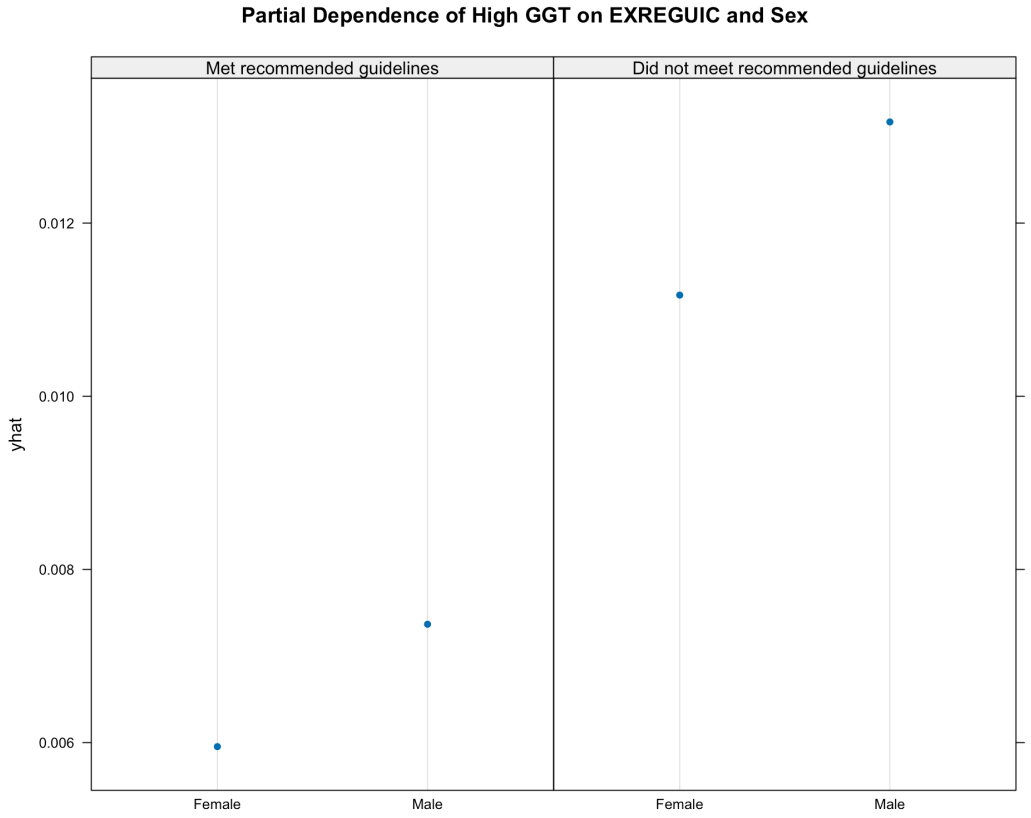
▼ Code

```
pdp.ex.age.ggt = partial(ggt.rf, pred.var = c("AGEB", "EXREGUIC'
                         train = fdat.full, which.class = "Hig
plotPartial(pdp.ex.age.ggt, contour = TRUE,
            main = "Partial Dependence of High GGT on EXREGUIC a
```



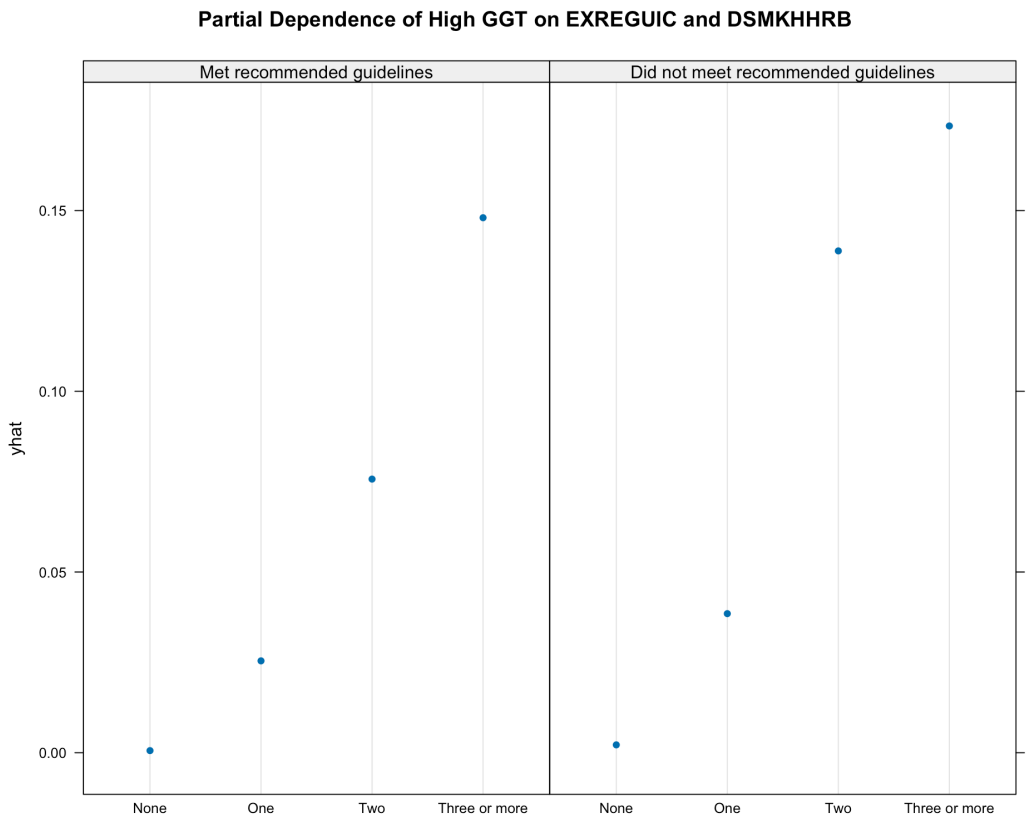
▼ Code

```
pdp.ex.sex.ggt = partial(ggt.rf, pred.var = c("SEX", "EXREGUIC"
                         train = fdat.full, which.class = "Hig
plotPartial(pdp.ex.sex.ggt, contour = TRUE,
            main = "Partial Dependence of High GGT on EXREGUIC a
```



▼ Code

```
pdp.ex.smoking.ggt = partial(ggt.rf, pred.var = c("DSMKHHRB", "
                             train = fdat.full, which.class = "Hi
plotPartial(pdp.ex.smoking.ggt, contour = TRUE,
            main = "Partial Dependence of High GGT on EXREGUIC a
```


Partial Dependence of High GGT on EXREGUIC and DSMKHHRB
Met recommended guidelines
Did not meet recommended guidelines
yhat
0.15
0.10
0.05
0.00
None
One
Two
Three or more
None
One
Two
Three or more